# Building Industry-Ready Skills with Model-Based Design, Simulink, and Agentic AI




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Education Application Engineer



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Education Application Engineer



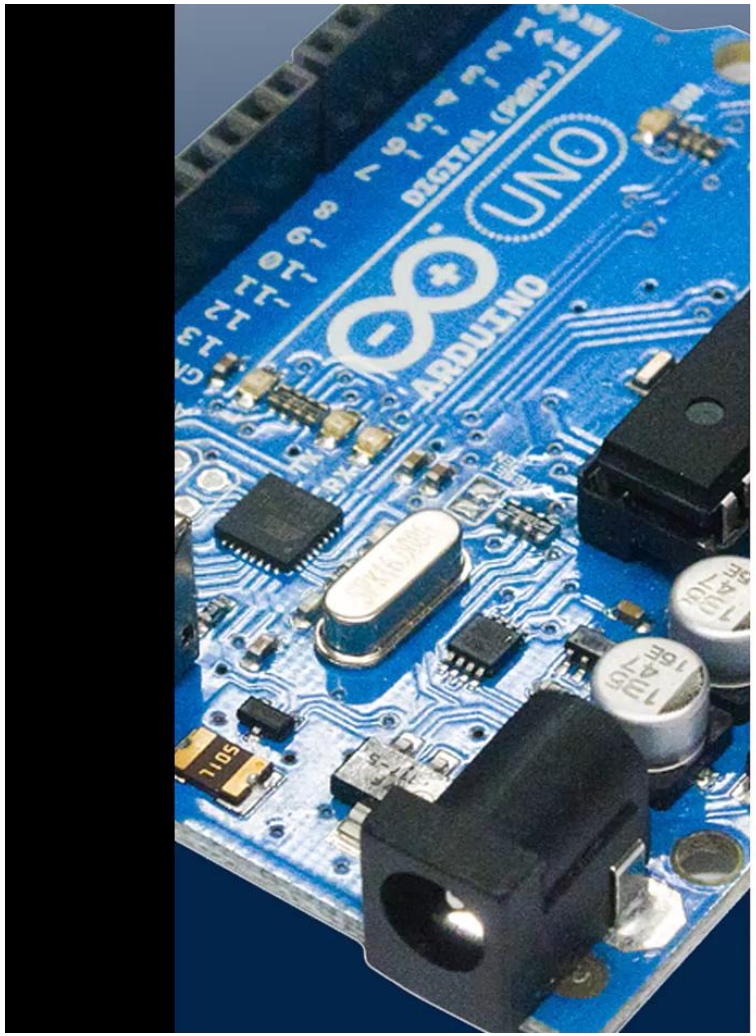

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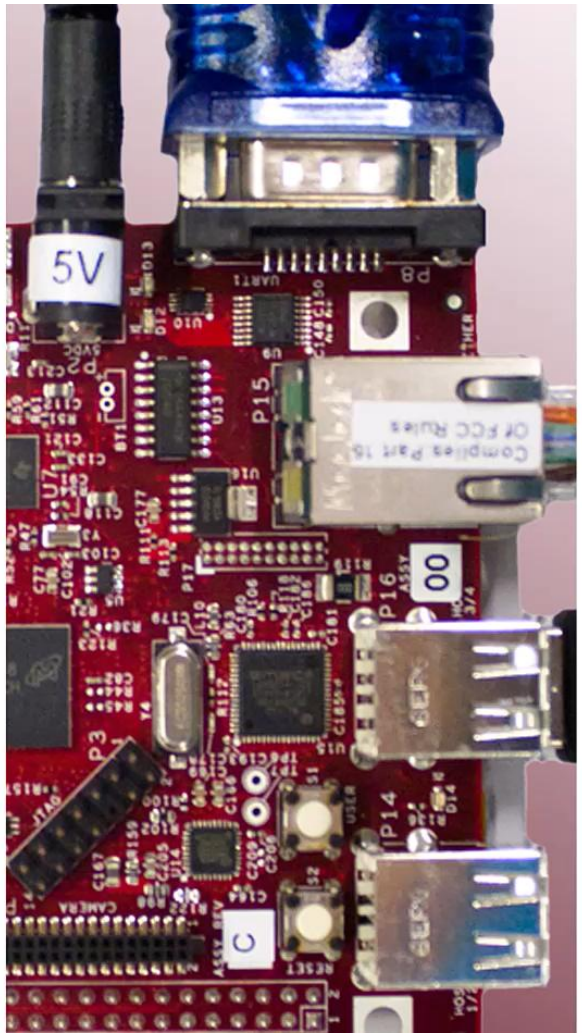
# To follow along

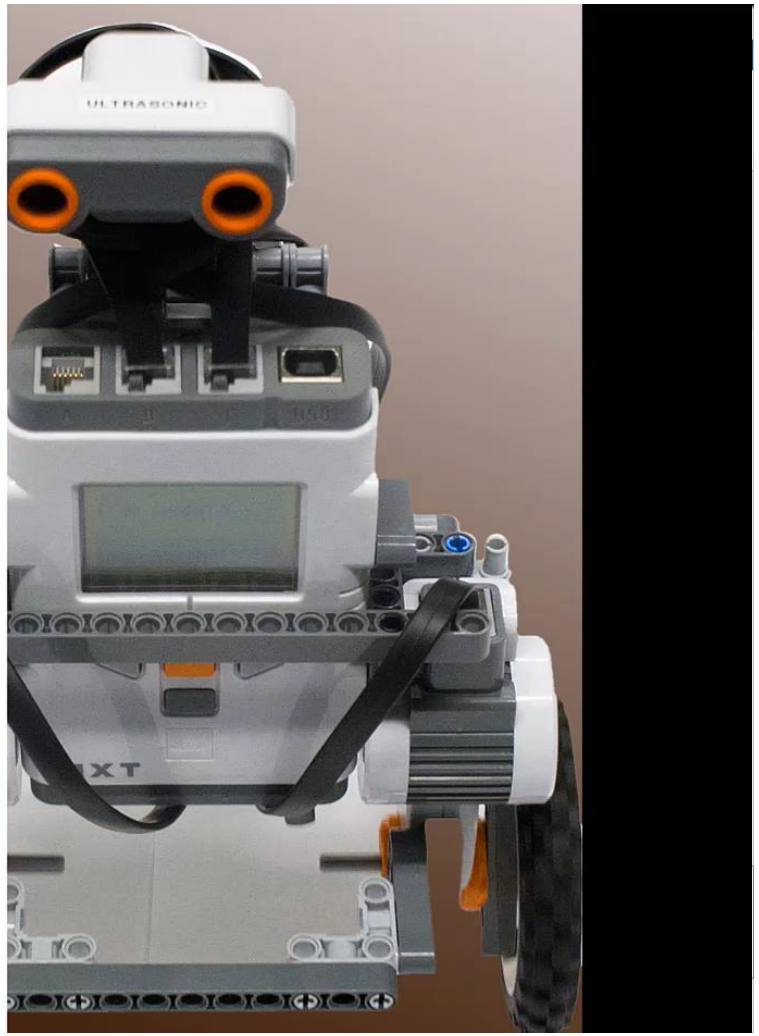
- Set up a MathWorks account if you don't have one
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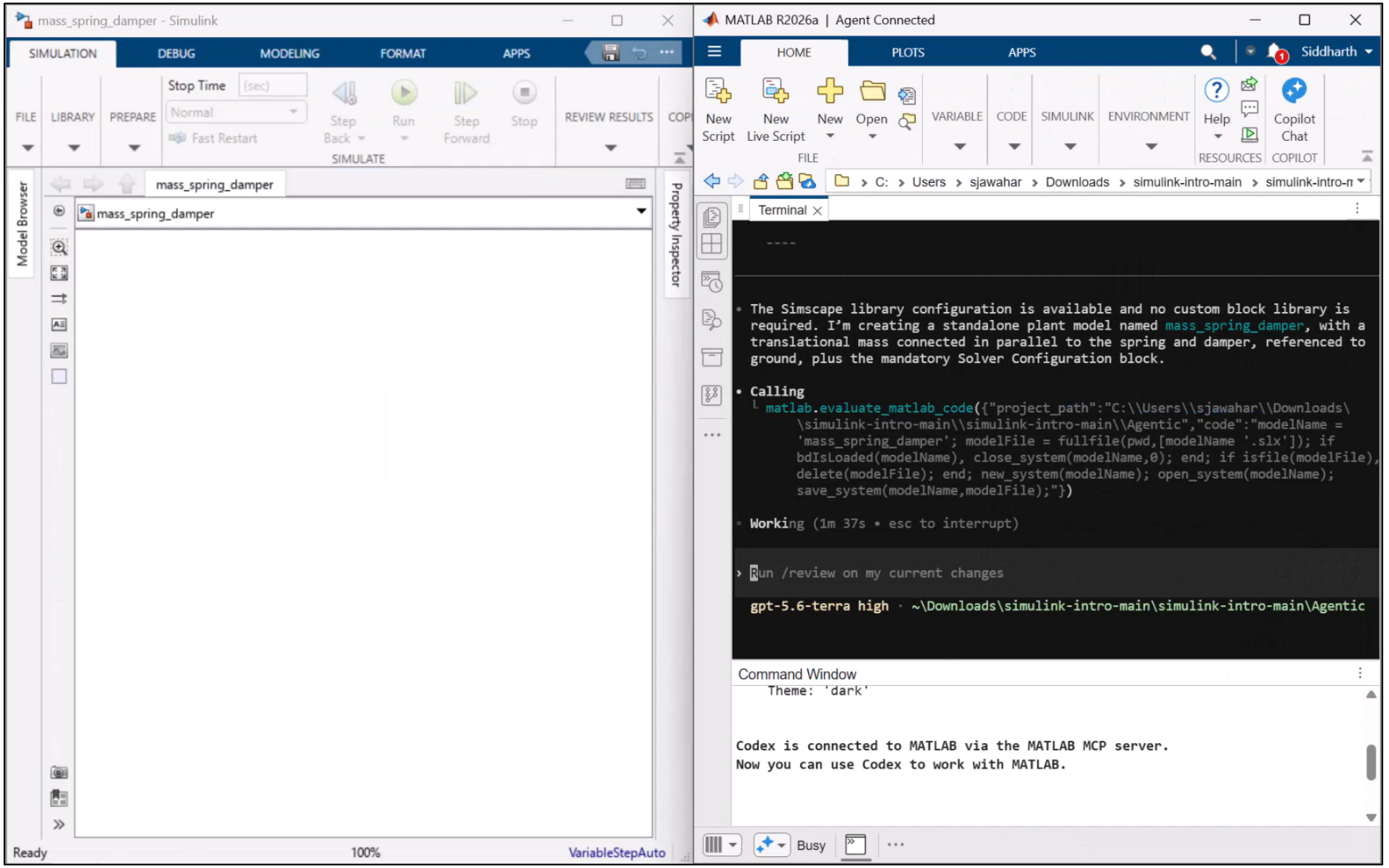
# What are we going to see today?



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# Agenda

- Introduction to Model-Based Design
- Simulink 101 – An Industry Standard tool for Model-Based Design
- AI use in Industry and Safety Critical Systems
- Resources

# Agenda

- **Introduction to Model-Based Design**
- Simulink 101 – An Industry Standard tool for Model-Based Design
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- Resources

# Let’s take a few examples of engineered systems…

Renewable Energy



Wireless Communication



Aerial Robotics



Electric Vehicles



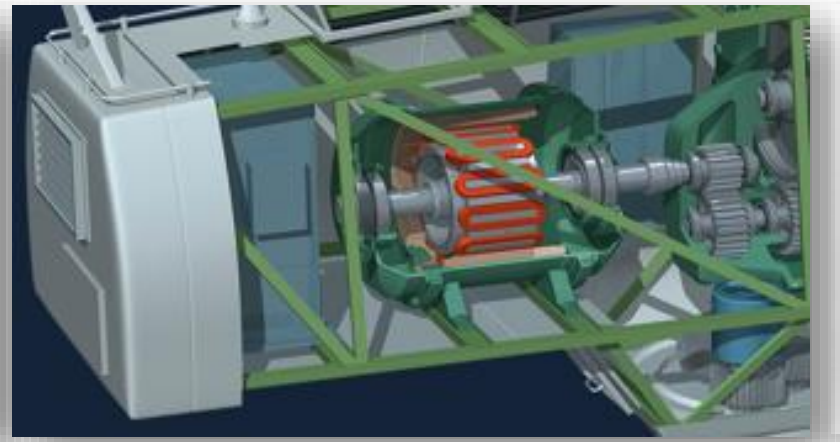
# Engineered systems require system-level, multi-domain integration

Renewable Energy

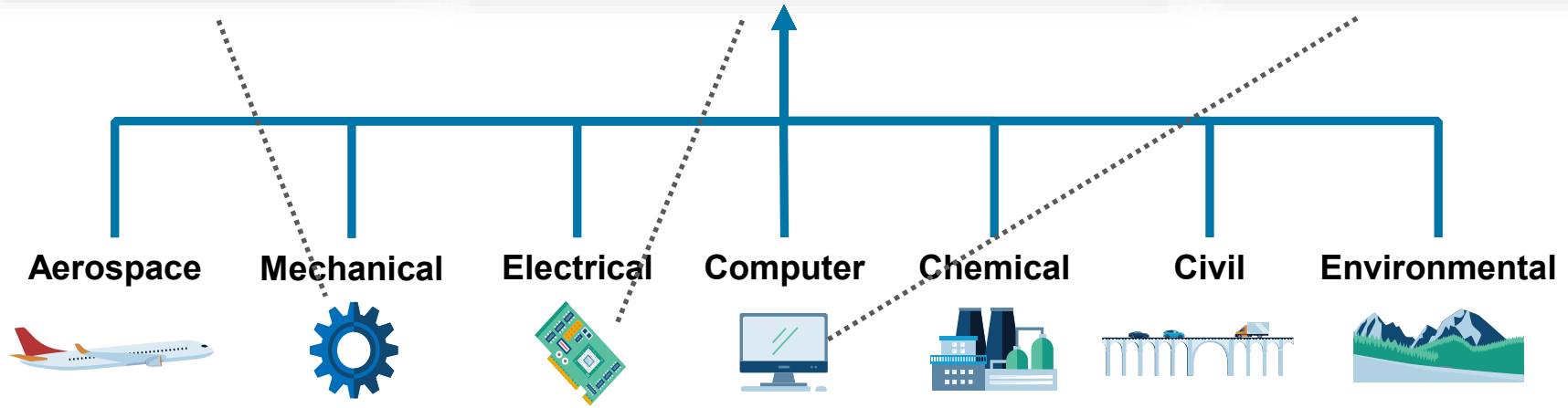




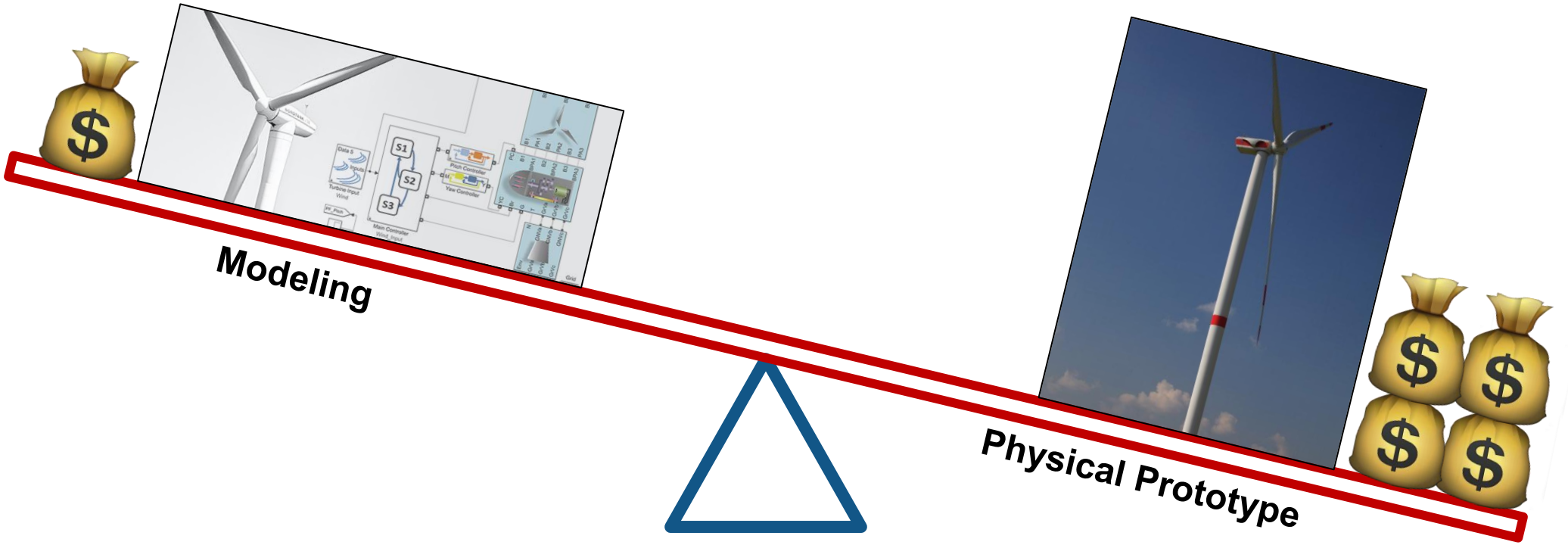








# Traditional development processes involving **physical prototypes** are **costly** and iterate **slowly**

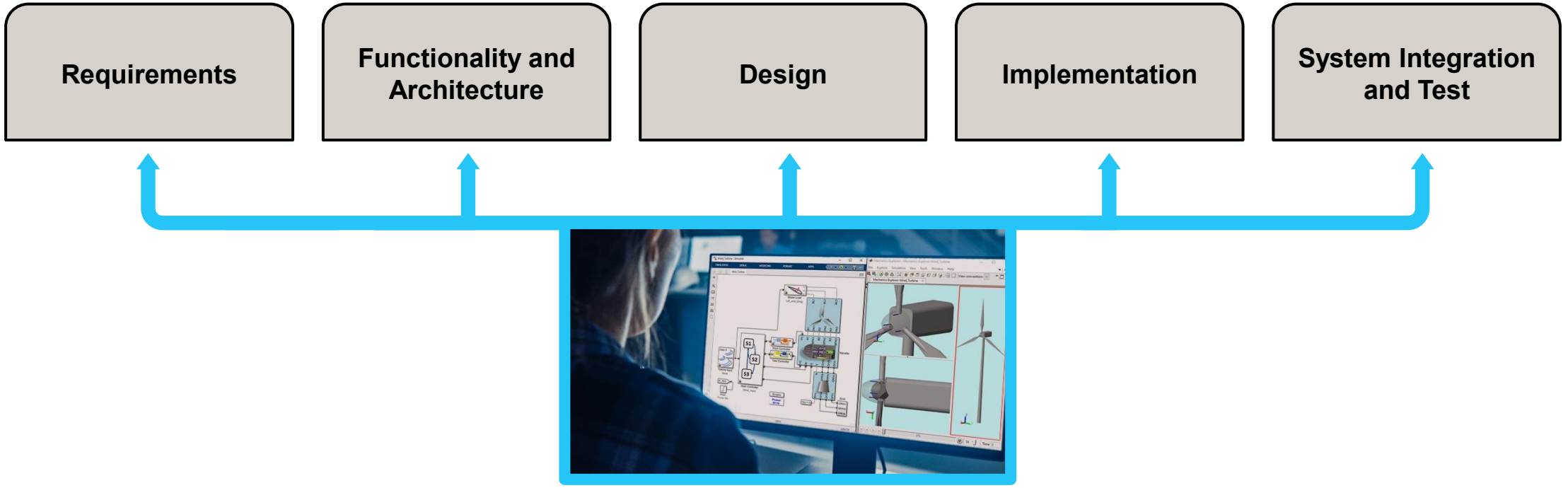



Design space **exploration**
Continuous design **improvement**

**Costly** and **time-consuming** to build
**Hinders** rapid iterations

# What is Model-Based Design?

Systematic use of **models** throughout the development process



# What is Model-Based Design?

Systematic use of **models** throughout the development process

**Modeling**



**Automation**

# What is Model-Based Design?

Systematic use of **models** throughout the development process

**Modeling**

Simulation | Analysis

+

**Automation**



# What is Model-Based Design?

Systematic use of **models** throughout the development process

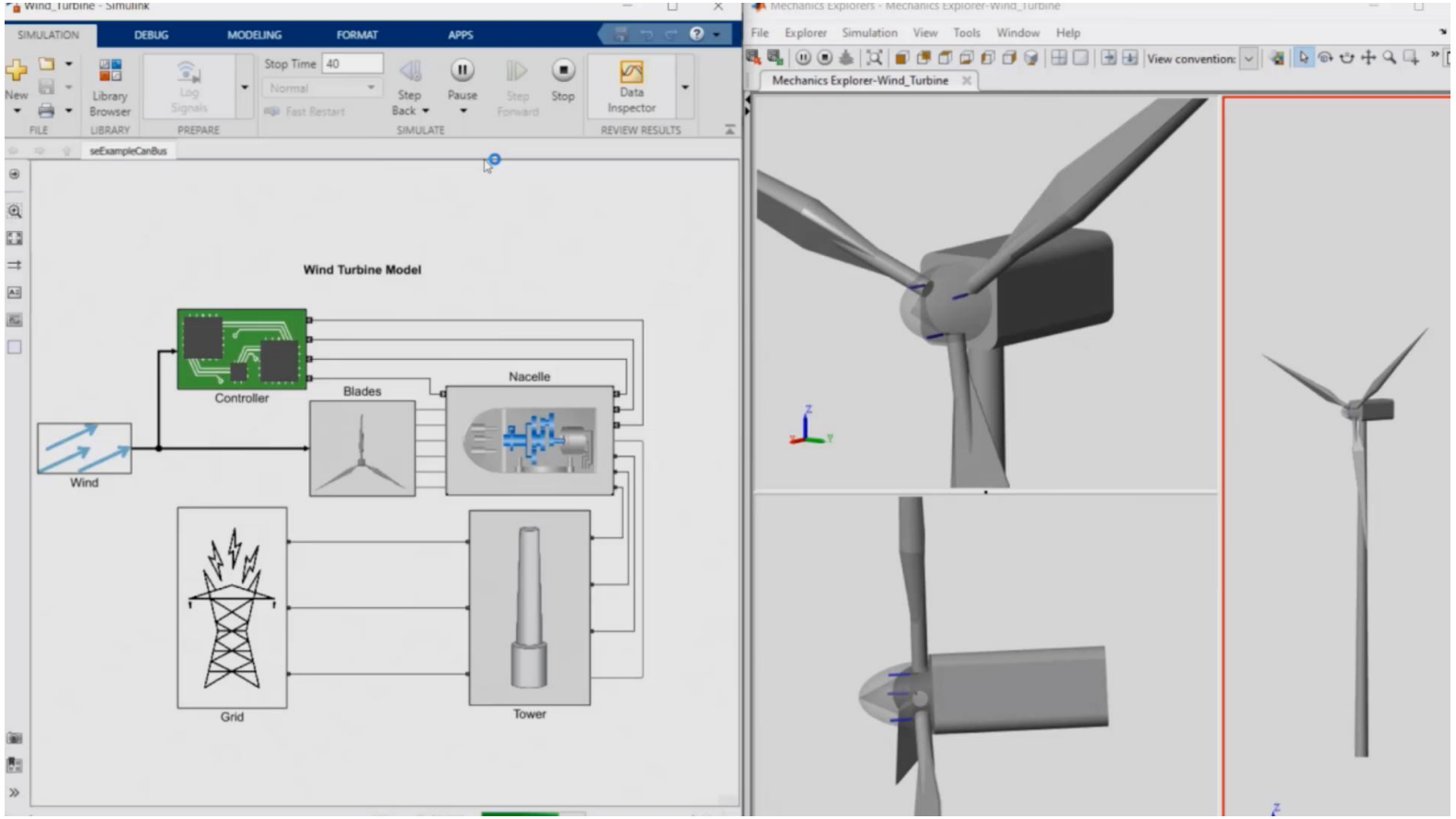
**Modeling**

Simulation | Analysis

+

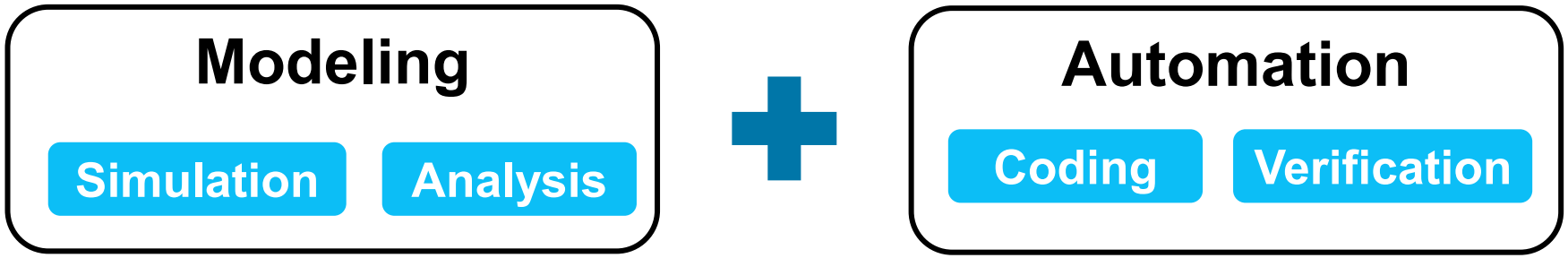
**Automation**

Coding | Verification



# Model-Based Design

Systematic use of **models** throughout the development process

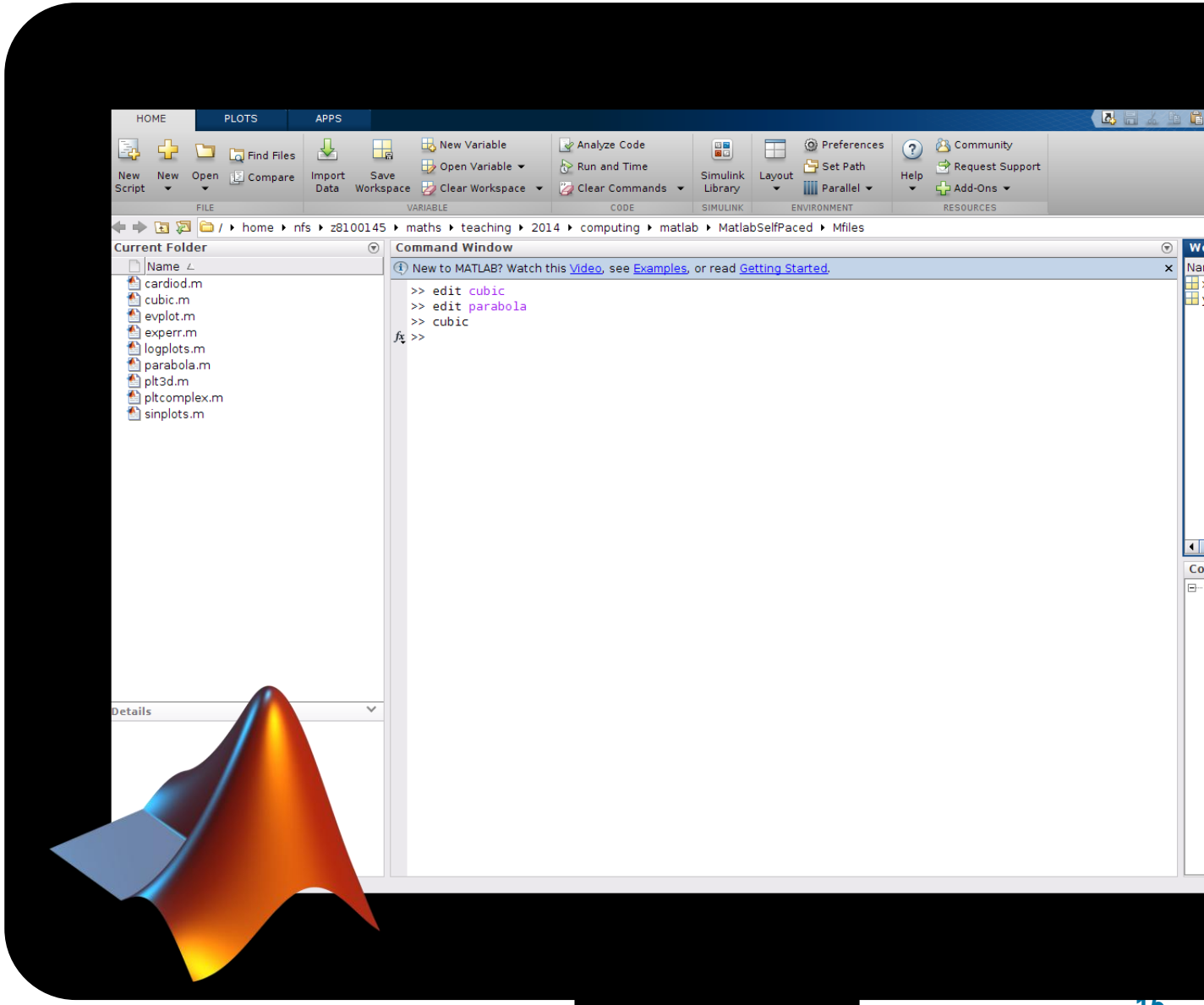



**Try out new ideas**
**Fast repeatable tests**

**Eliminate manual steps**
and **human error**

# MATLAB and Simulink

Programming platform designed specifically for algorithm development, data analysis, visualization, and numeric computation.




## What Can I Do With MATLAB?

- Analyze data
- Develop algorithms
- Create AI models and applications

MATLAB lets you take your ideas from research to production, as well as integrating with Simulink® and Model-Based Design.

Over 100 add-on products for specialized tasks are available.

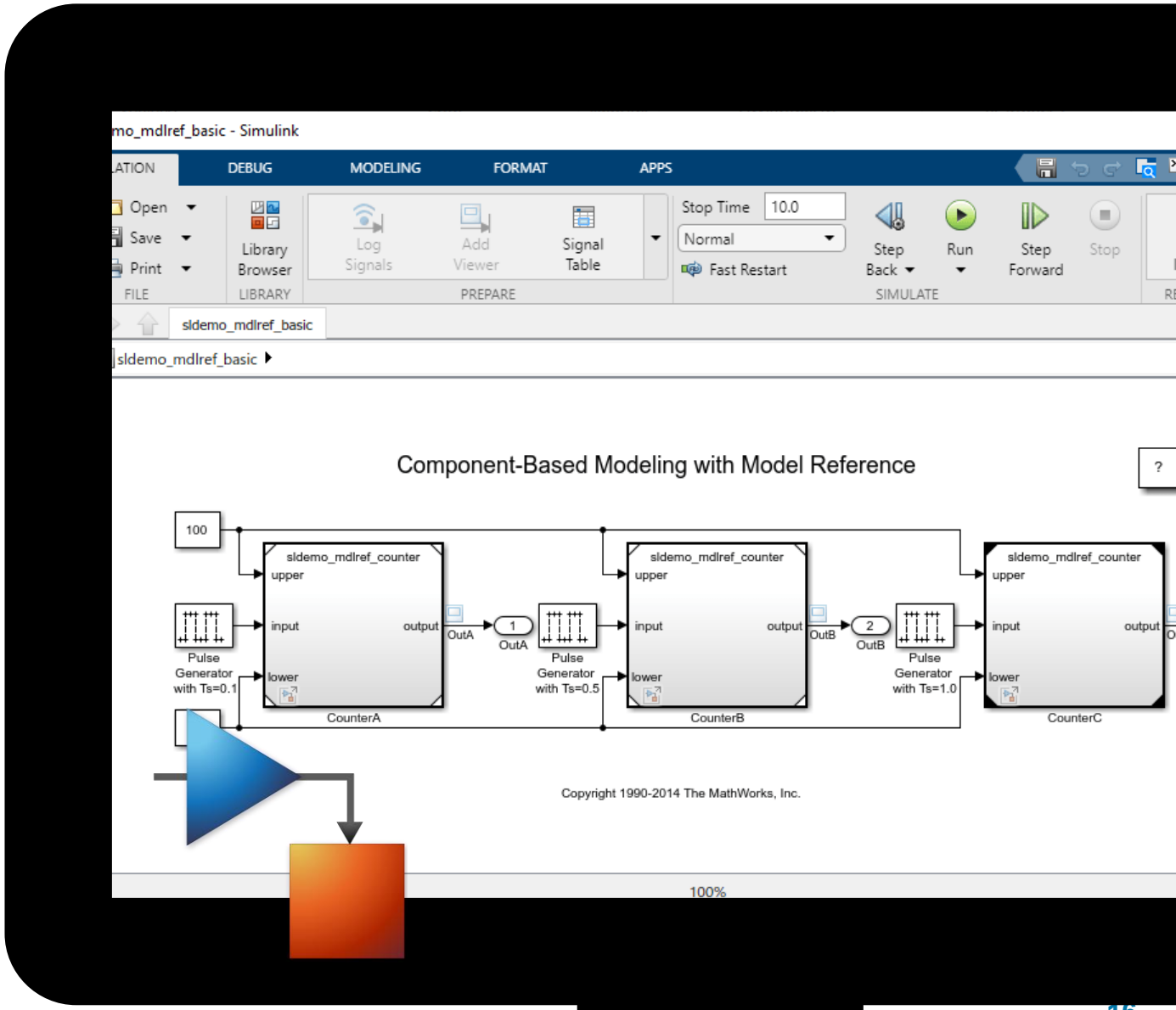
# MATLAB and Simulink

A block diagram environment used to design systems with multidomain models, simulate before moving to hardware, and deploy without writing code..



## Simulink is for *Model-Based Design*

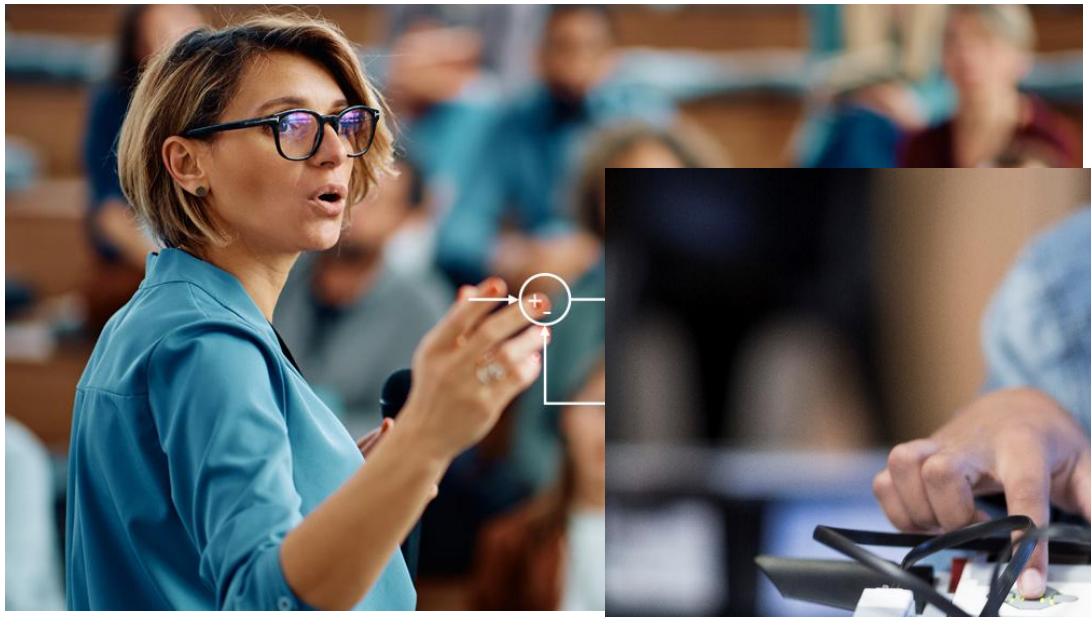
- Use a virtual model to simulate and test your system early and often
- Validate your design
- Design and simulate your system before moving to hardware



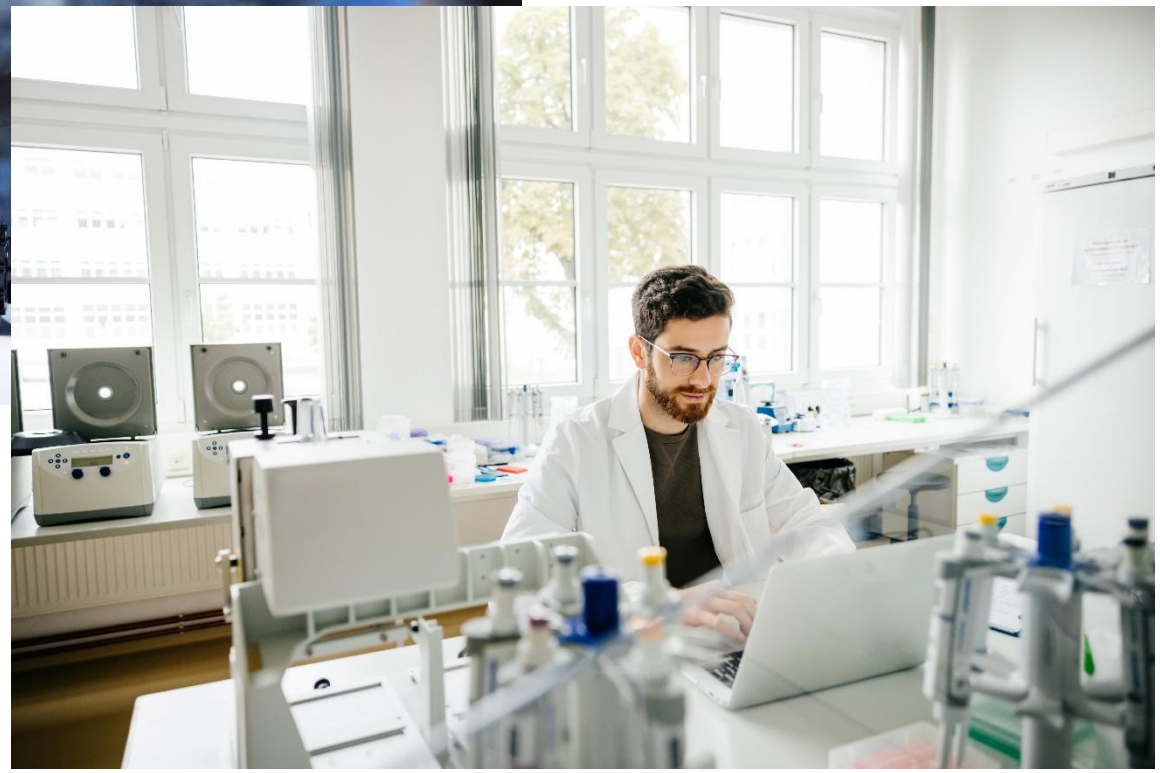
# End to end advanced air mobility development at Supernal

*“Complete flight simulation of autonomous vehicle operation using photorealistic environment”*









# TAM Perspective: Industry roles and skill areas

**What is a TAM (Technical Account Manager)?**

- Technical expert in MathWorks tools
- Technical / business expert in Industry

**How do we work with Industry customers?**

- Identifying customer corporate goals
- Developing strategies to support those goals using our tools and methods

**What kinds of groups do we interact with / use our tools?**

- Engineering for Product Development
- Research & Development
- Data Analytics
- Finance

# Industry insider view



## Hiring Manager Perspective

1 in 10 engineers had the right skills in MBD

Find industry contacts and use them!

Training is available for campus-wide licenses

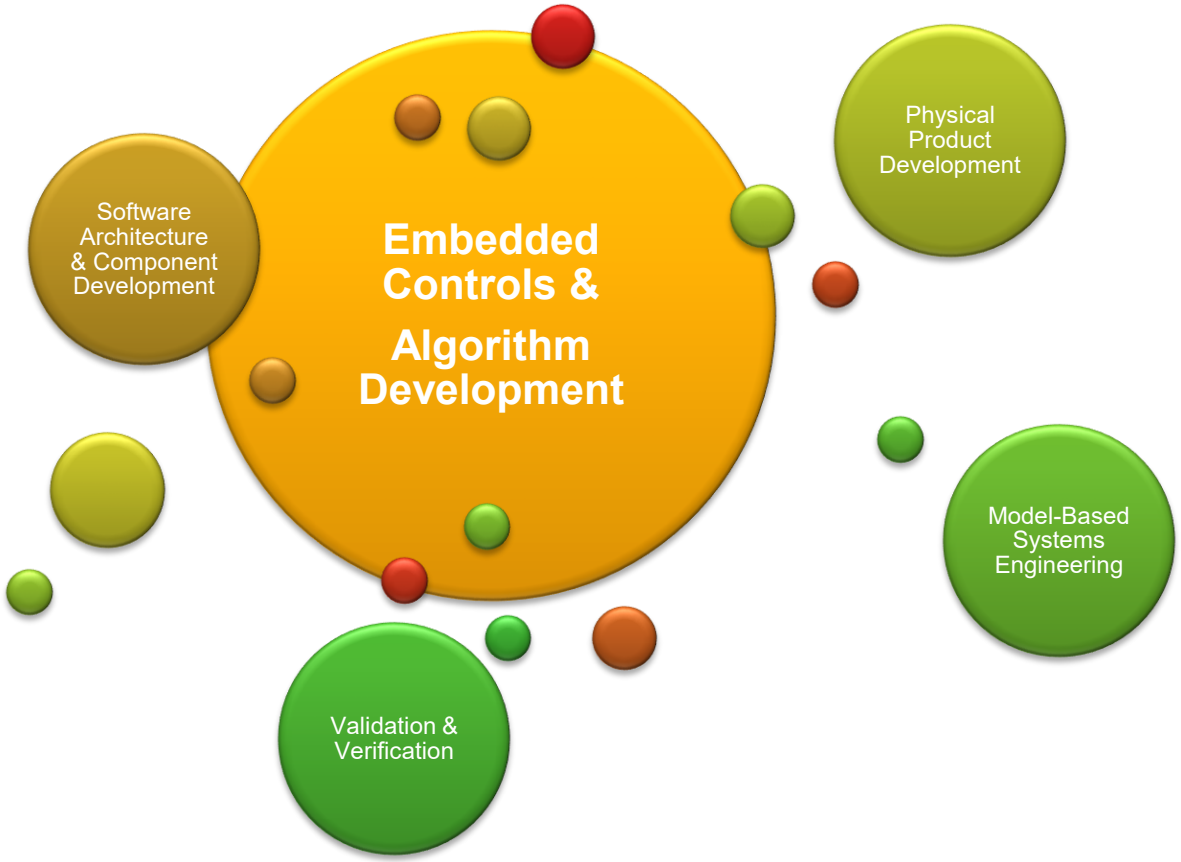


## Internships

Show direct industry experience directly to hiring managers

Search with keywords: “Model-Based Design”, “Simulink”, and “Code Generation”

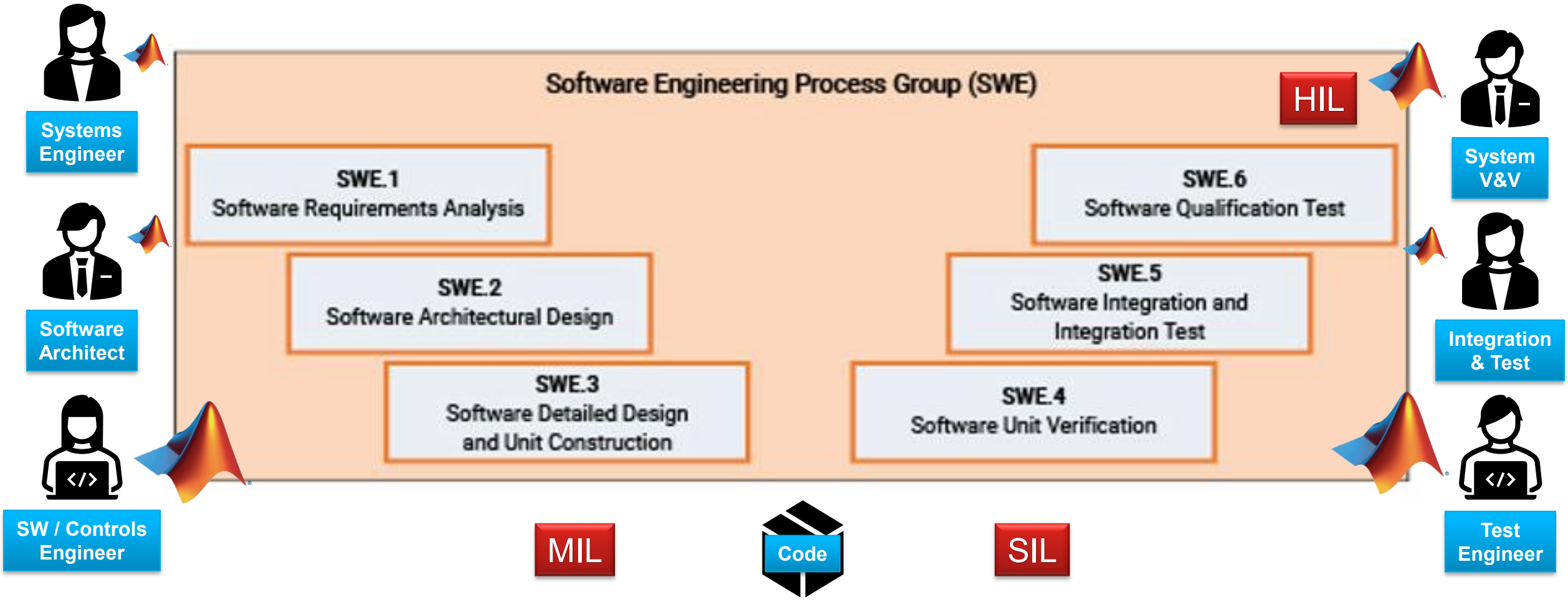
# Industry roles for Model-Based Design




- Embedded Controls Development
  - Controller Modeling
  - Software Component Modeling
  - Requirements-Based Model-In-the-Loop (MIL) Testing
  - Production Code Generation
  - Software-In-the-Loop (SIL) Testing
  - Integration Testing
  - Hardware-In-the-Loop (HIL) Testing
- How many engineering positions?
  - Large companies: ~1000s

# MATLAB and Simulink in product development workflows

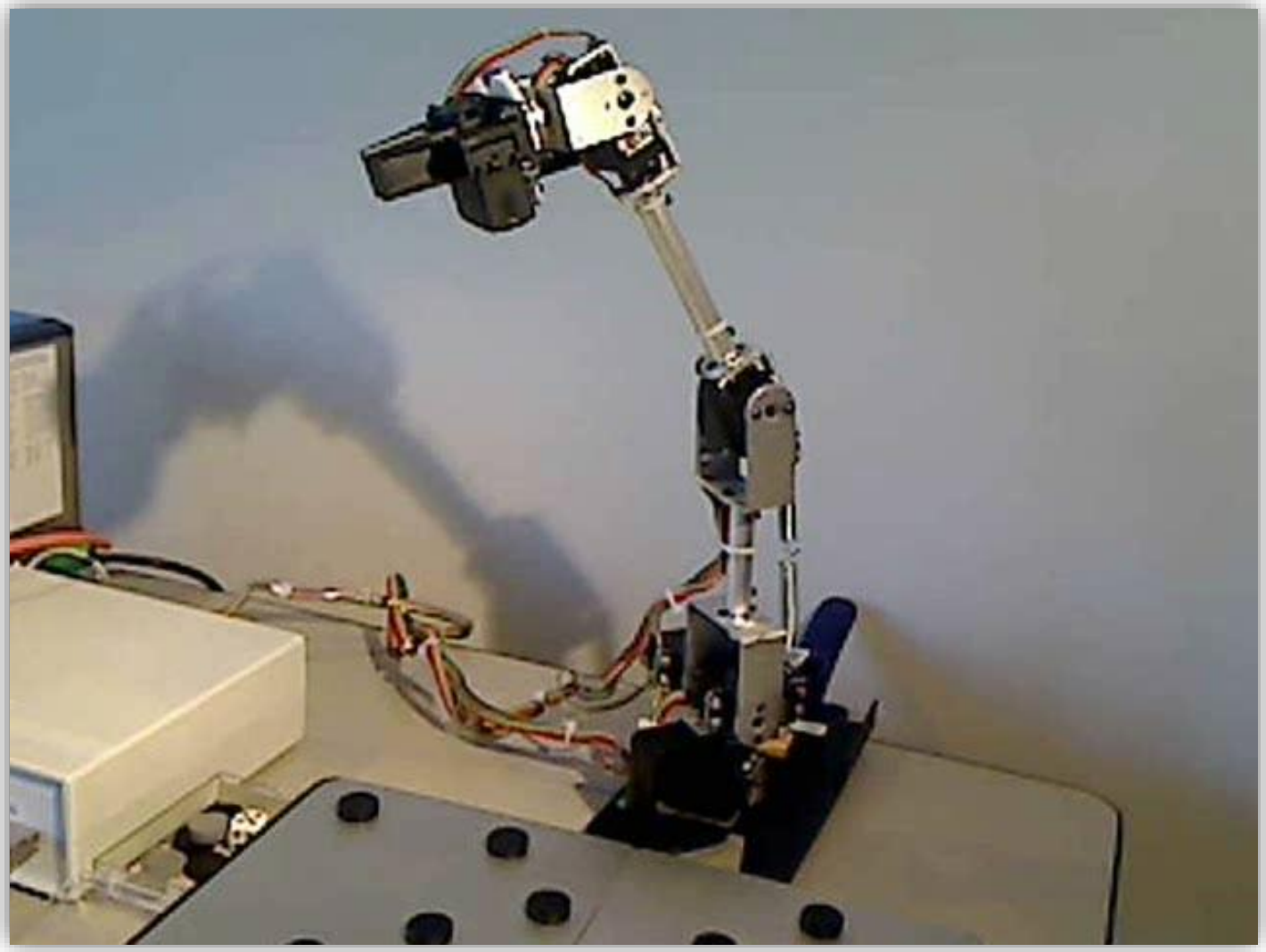
## Example process for safety-critical software



# Agenda

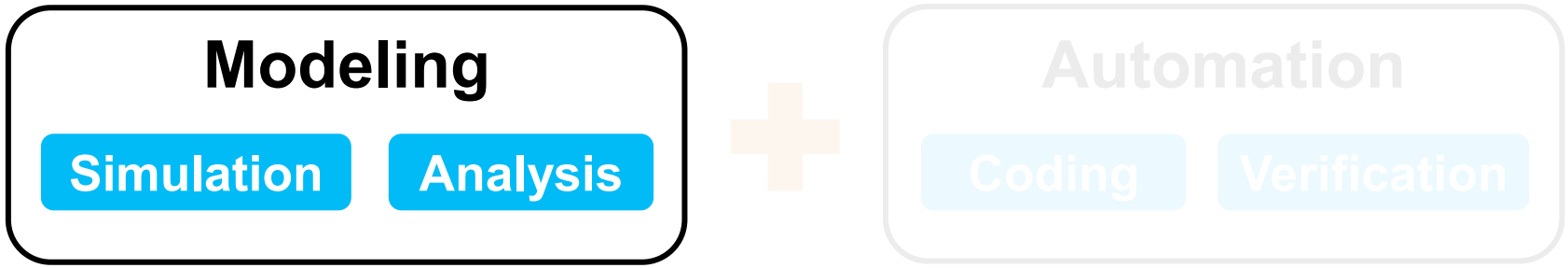
- Introduction to Model-Based Design
- **Simulink 101 – An Industry Standard tool for Model-Based Design**
- AI use in Industry and Safety Critical Systems
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# Let’s look at a simple pick and place robot…

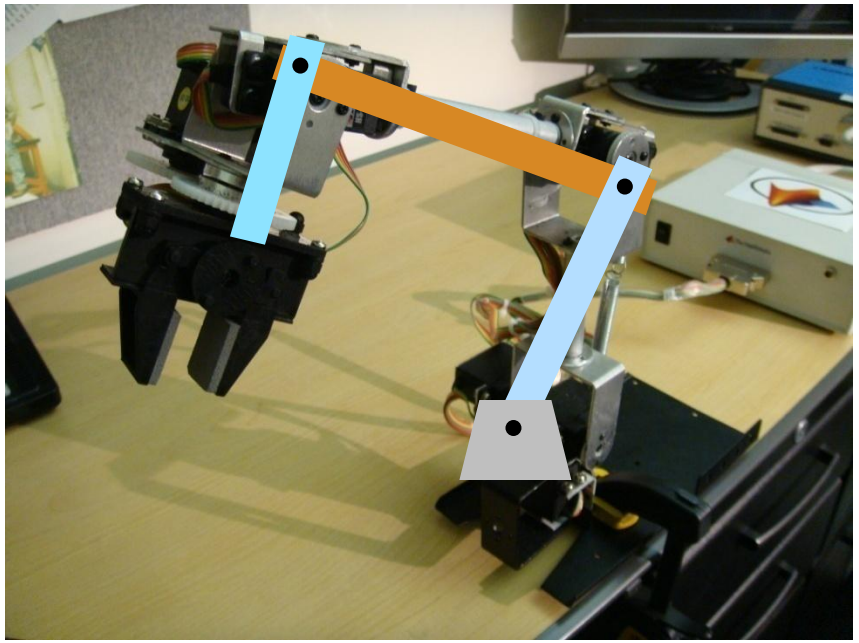


# Model-Based Design

Systematic use of **models** throughout the development process



# To model the system, let's break it down into its subcomponents

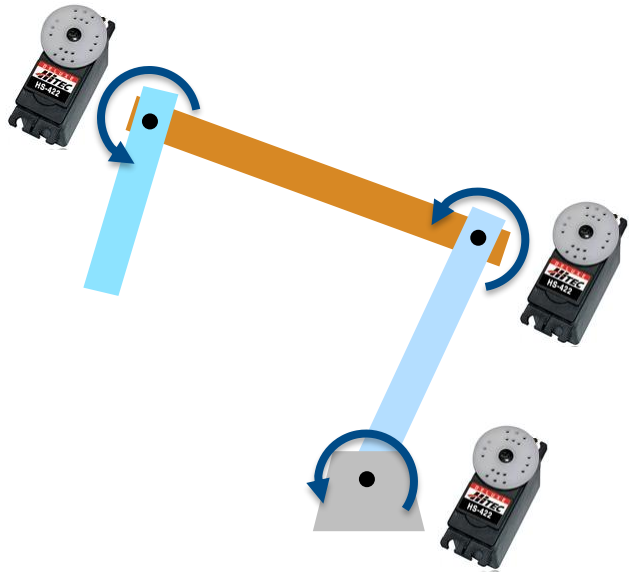


Modeling

Simulation | Analysis

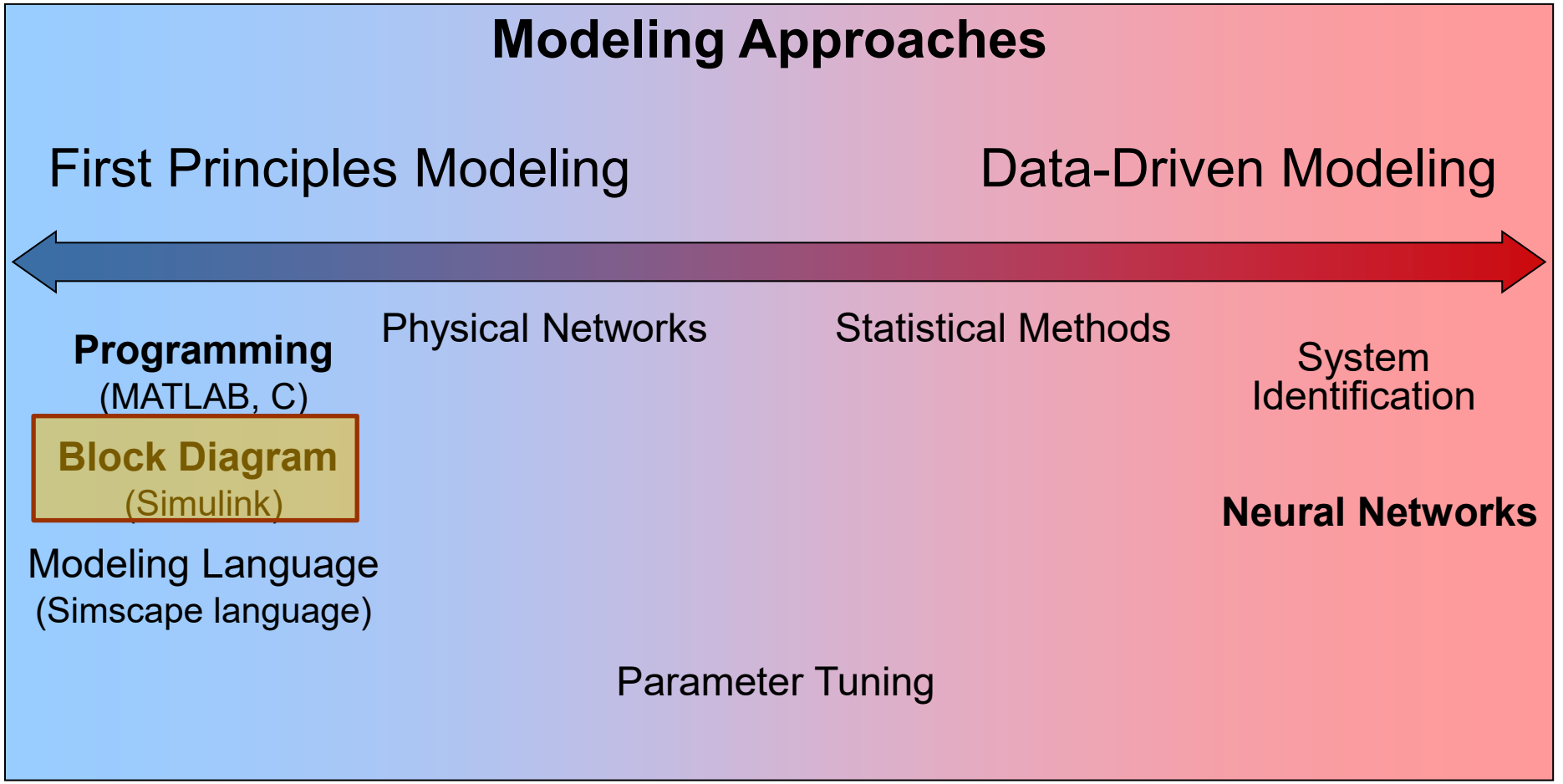
# To model the system, let's break it down into its subcomponents



**Let's model the DC motor!**

# There are several approach to modeling dynamic systems




Modeling

Simulation | Analysis

# We know the governing equations for a DC motor



$$V = K\omega + iR + L\frac{di}{dt}$$

$$\frac{di}{dt} = \frac{1}{L}(V - K\omega - iR)$$

$$i = \int \frac{1}{L}(V - K\omega - iR)dt$$

*Electrical*

$$J\frac{d\omega}{dt} = Ki - b\omega - T_{Load}$$

$$\frac{d\omega}{dt} = \frac{1}{J}(Ki - b\omega - T_{Load})$$

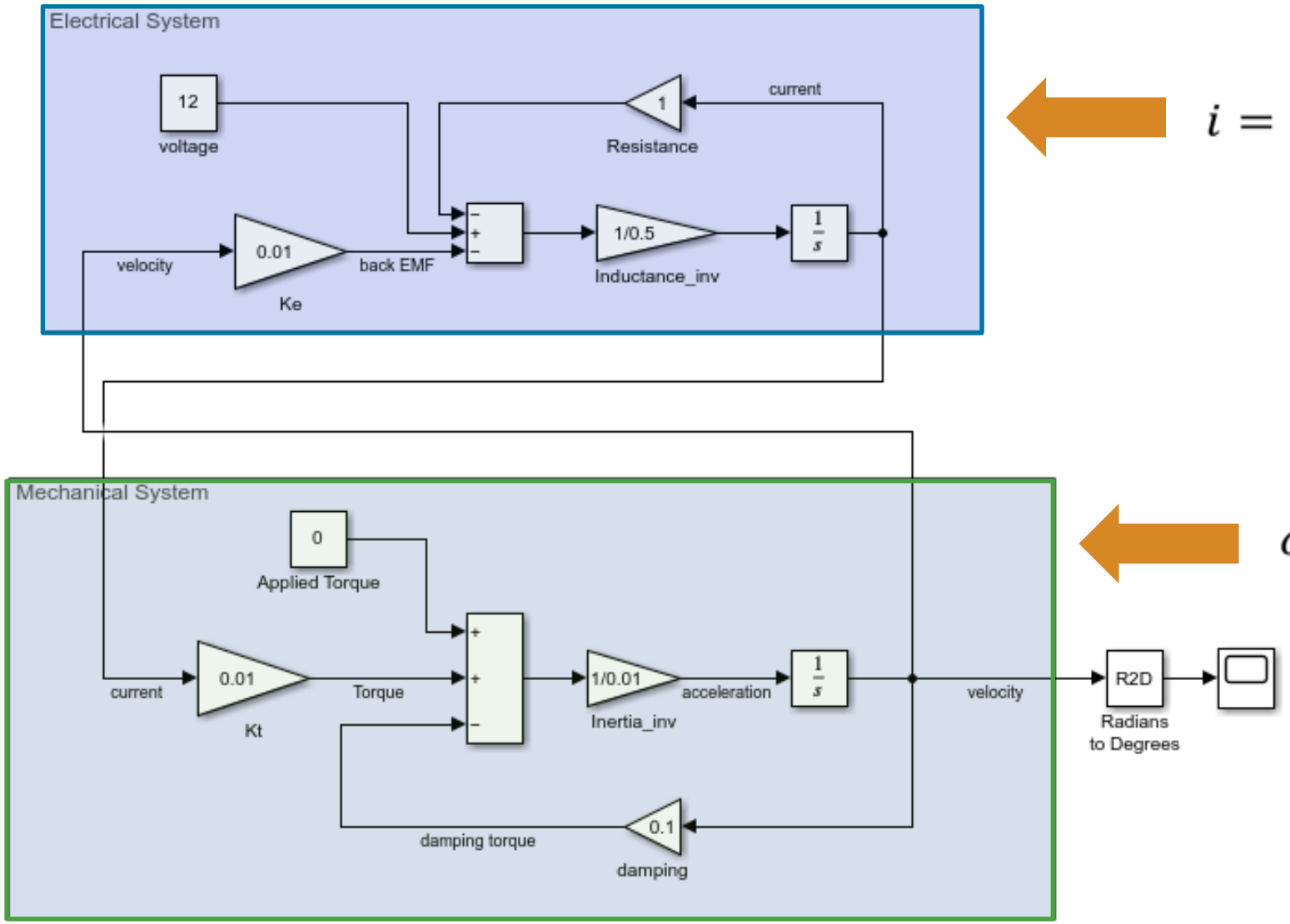
$$\omega = \int \frac{1}{J}(Ki - b\omega - T_{Load})dt$$

*Mechanical*

**Modeling**

**Simulation** **Analysis**

# Let's model these equations in Simulink




$$i = \int \frac{1}{L}(V - K\omega - iR)dt$$

*Electrical*

$$\omega = \int \frac{1}{J}(Ki - b\omega - T_{Load})dt$$

*Mechanical*

**Modeling**

Simulation | Analysis

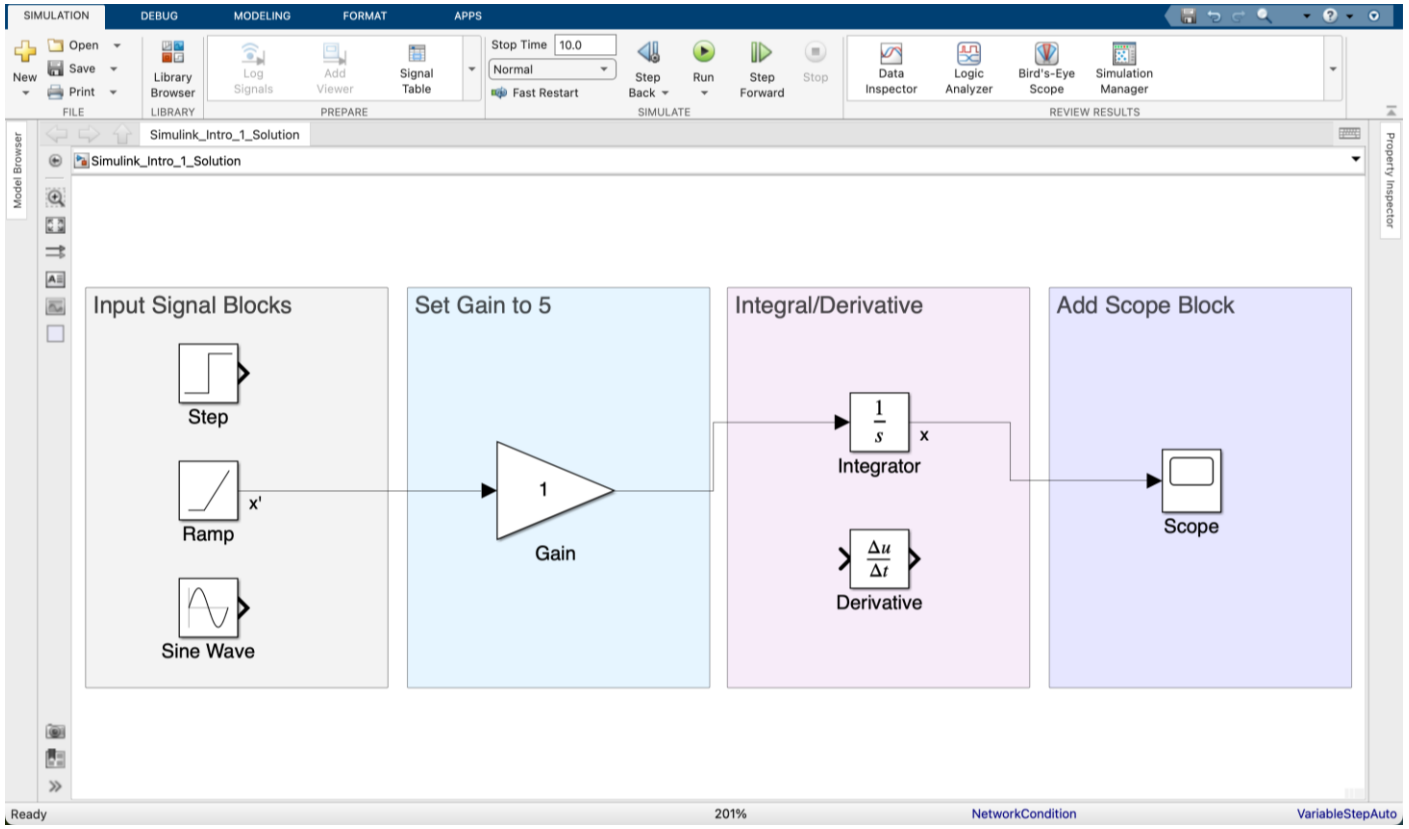
# To follow along

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  - go to https://www.mathworks.com/mwaccount/
  - Then open up MATLAB Online (https://matlab.mathworks.com/)
- Copy the materials via GitHub
  - https://github.com/SidJawMW/simulink-intro



# Exercise 1: Introduction to Simulink

- Open: **Simulink_Intro.slx**
- Connect Input blocks to Gain and Integral/Derivative blocks
- Add a Scope block and connect signals to view them
- Adjust Stop Time to change simulation duration

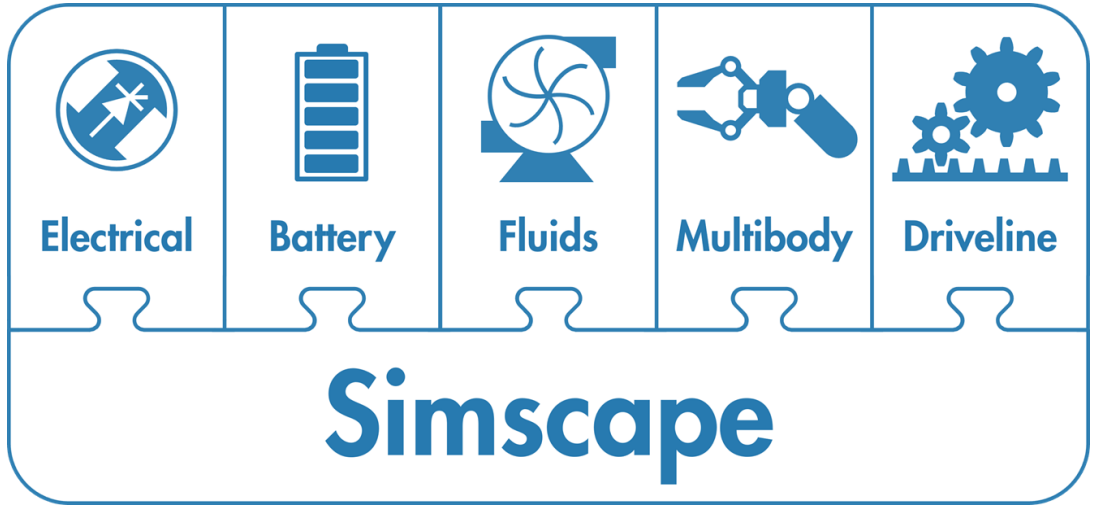


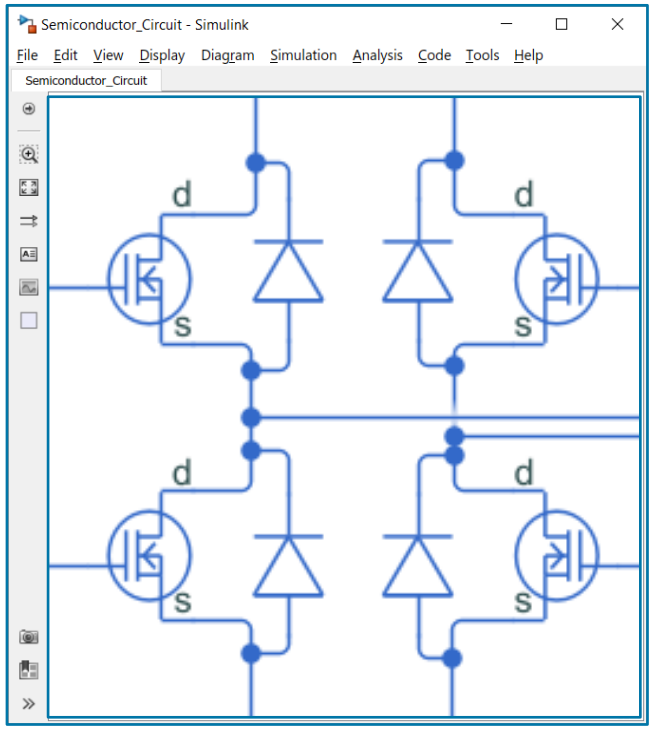

**Modeling**

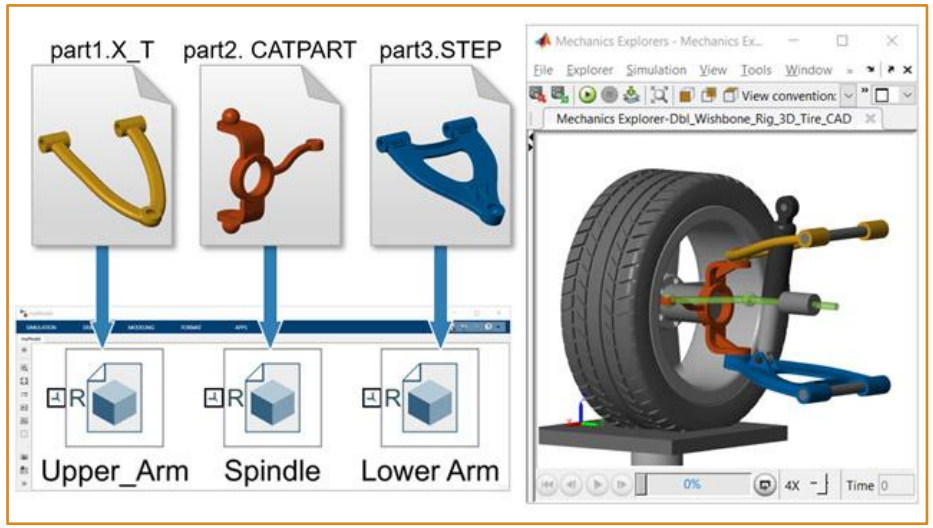
Simulation | Analysis

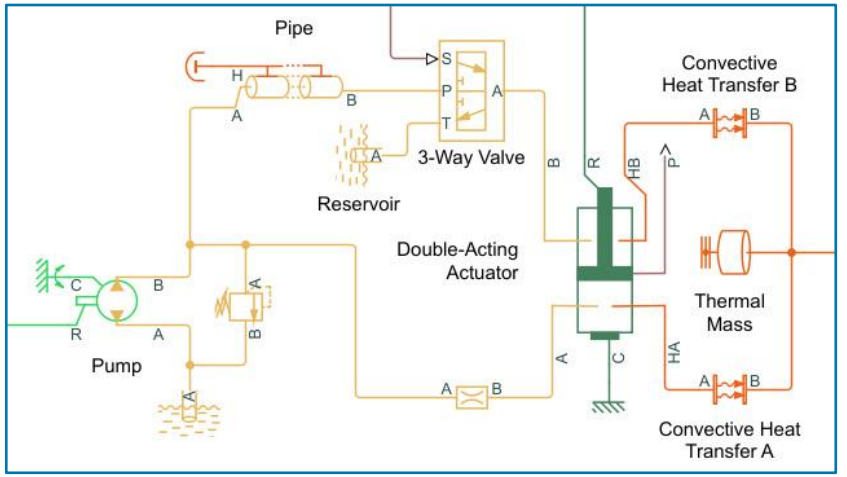
# Simscape
## Overview

- Simscape enables you to rapidly create models of physical systems within the Simulink environment

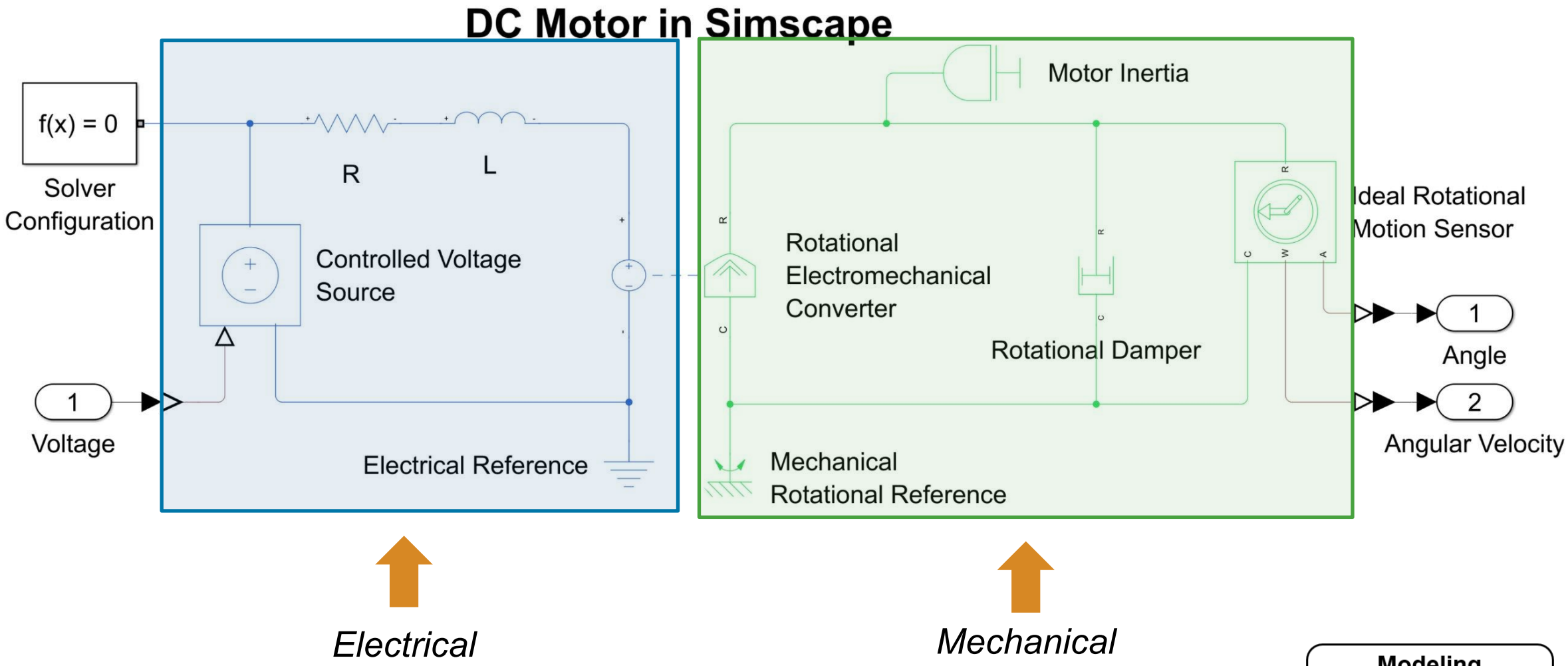


# Physical network model of a DC motor using Simscape

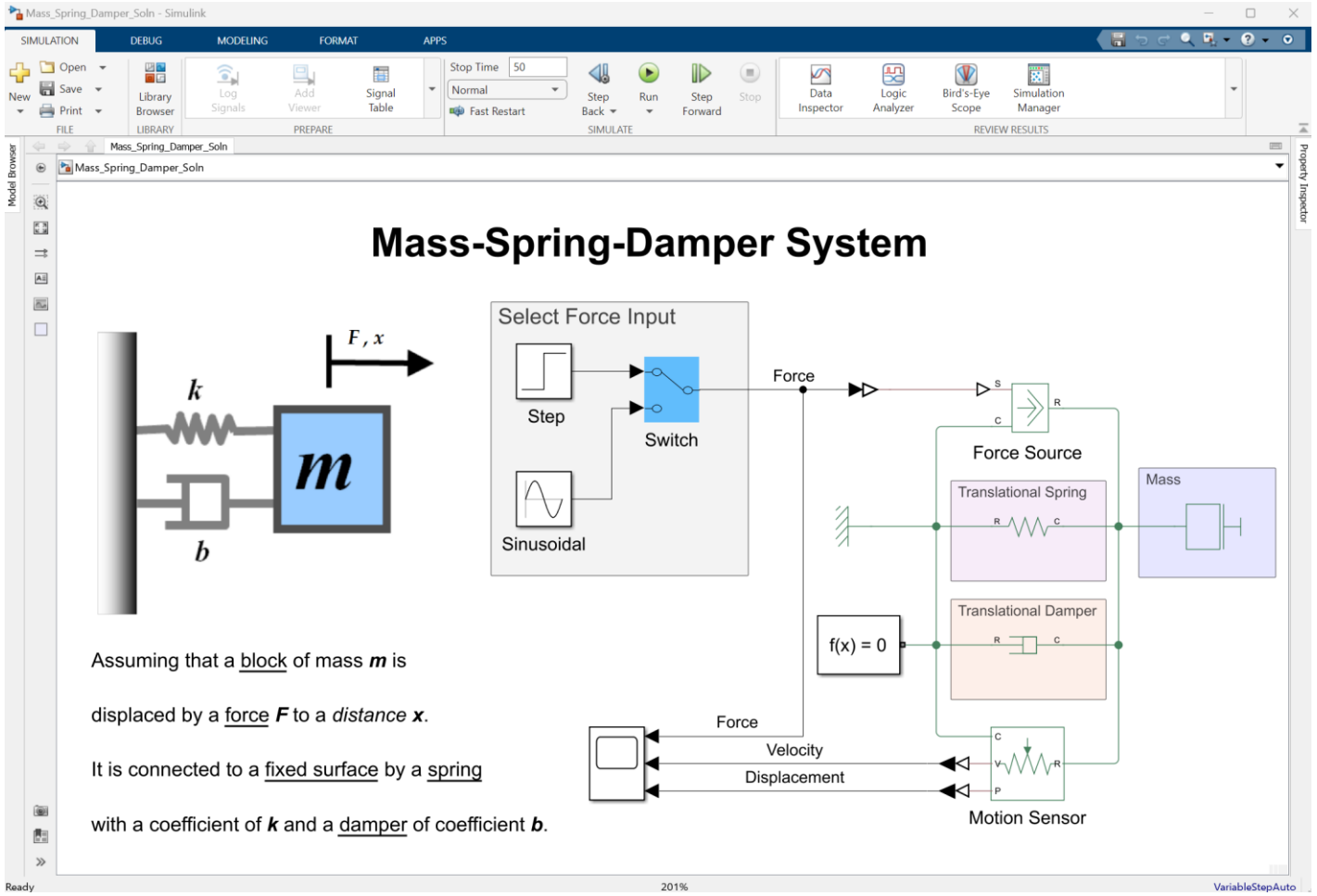



Modeling

Simulation | Analysis

# Exercise 2: Mass-Spring Damper

- Open: ***Mass_Spring_Damper.slx***
- Add Translational Spring, Translational Damper, and Mass blocks
- Edit block parameters and view results in Scope block




**Modeling**

**Simulation** **Analysis**

# MATLAB, Simulink, and Simscape for physical modeling

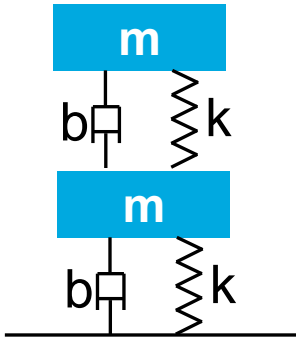
$$F_{Spring} = k_{Spring} * (x_{Mass})$$

$$F_{Damper} = b_{Damper} * \left(\frac{dx_{Mass}}{dt}\right)$$

$$\frac{d^2x_{Mass}}{dt^2} = \frac{-F_{Spring} - F_{Damper}}{m_{Mass}}$$

$$F_{Spring2} = k_{Spring} * (x_{Mass})$$

....




Below is a simple script using MATLAB code to simulate the dynamics of a spring-mass-damper system.

```
% Parameters
m = 1;
k = 10;
b = 1;

% Initial c
x0 = 1;
v0 = 0;

% Time span
tspan = [0

% ODE funct
odefun = @(

% Solve ODE
[t, y] = od

% Plot resu
figure;
subplot(2,
plot(t, y(:
title('Disp
xlabel('Tim
ylabel('Dis

subplot(2,
plot(t, y(:
title('Velo
xlabel('Tim
ylabel('Vel
```

Below is a simple script using MATLAB code to simulate the dynamics of a double spring-mass-damper system.

```
% Parameters
m1 = 1;      % mass 1 (kg)
m2 = 1;      % mass 2 (kg)
k1 = 10;     % spring constant 1 (N/m)
k2 = 10;     % spring constant 2 (N/m)
b1 = 1;      % damping coefficient 1 (Ns/m)
b2 = 1;      % damping coefficient 2 (Ns/m)

% Initial conditions
x1_0 = 1;    % initial displacement of mass 1 (m)
v1_0 = 0;    % initial velocity of mass 1 (m/s)
x2_0 = 0;    % initial displacement of mass 2 (m)
v2_0 = 0;    % initial velocity of mass 2 (m/s)

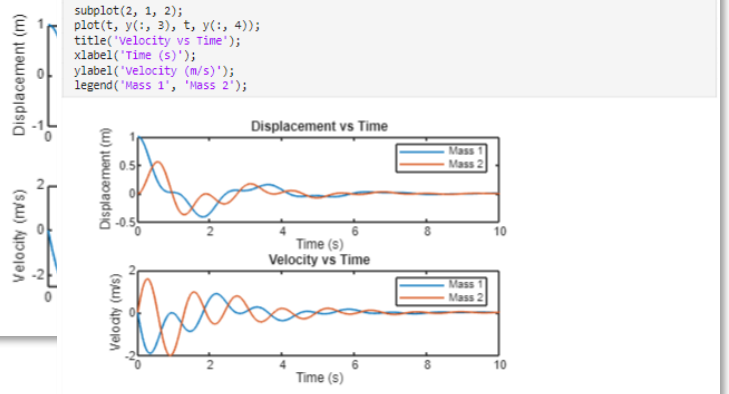
% Time span
tspan = [0 10]; % simulate from 0 to 10 seconds

% ODE function
odefun = @(t, y) [y(3); y(4);
                  (-b1 * y(3) - k1 * (y(1) - y(2))) / m1;
                  (-b2 * y(4) + k1 * (y(1) - y(2)) - k2 * y(2)) / m2];

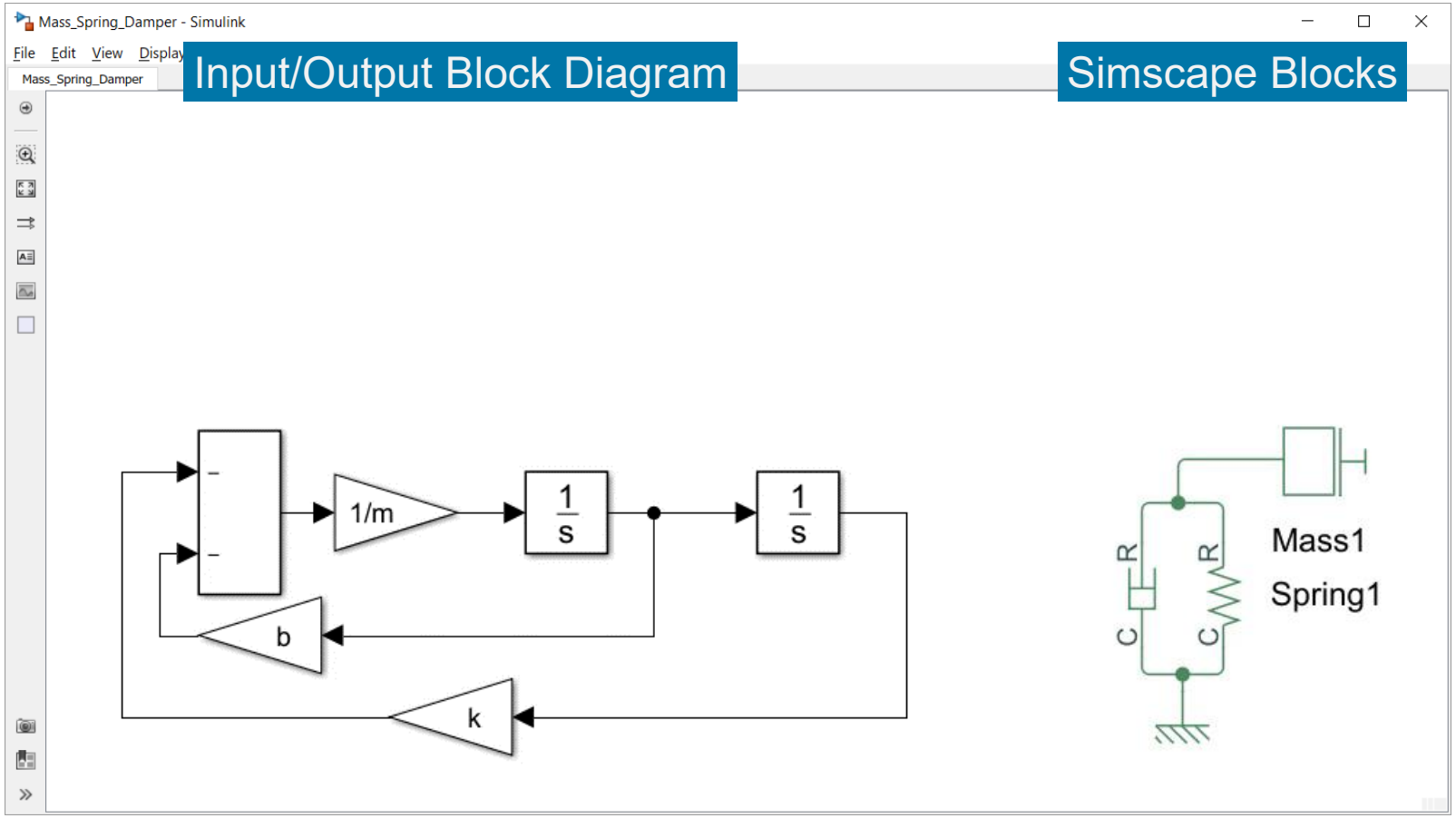
% Solve ODE
[t, y] = ode45(odefun, tspan, [x1_0; x2_0; v1_0; v2_0]);

% Plot results
figure;
subplot(2, 1, 1);
plot(t, y(:, 1), t, y(:, 2));
title('Displacement vs Time');
xlabel('Time (s)');
ylabel('Displacement (m)');
legend('Mass 1', 'Mass 2');

subplot(2, 1, 2);
plot(t, y(:, 3), t, y(:, 4));
title('Velocity vs Time');
xlabel('Time (s)');
ylabel('Velocity (m/s)');
legend('Mass 1', 'Mass 2');
```



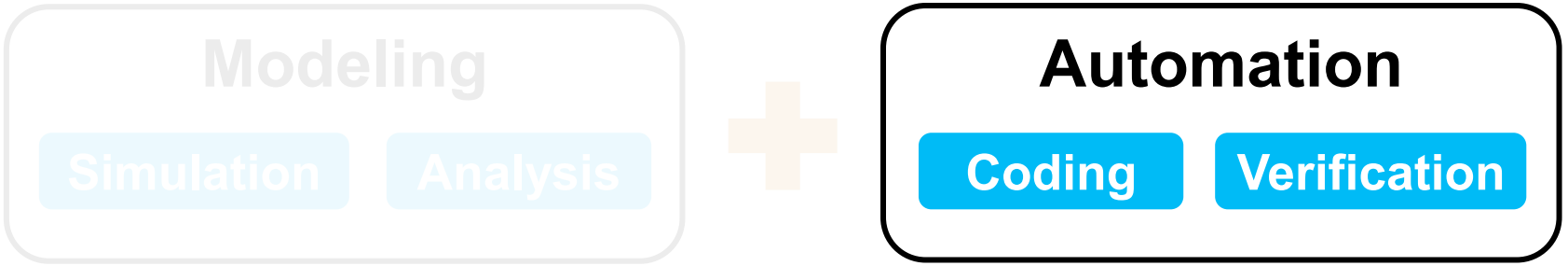

MATLAB



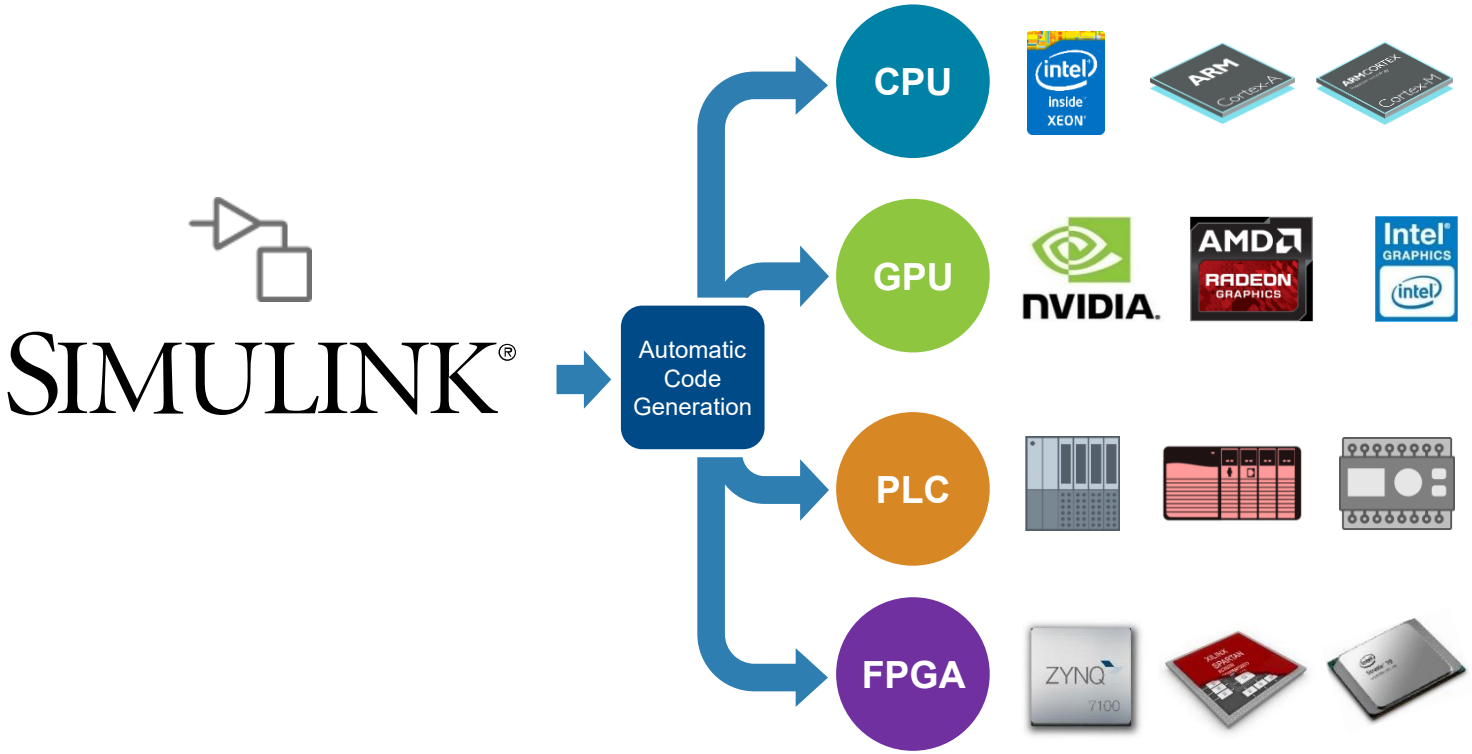

Simulink

# Model-Based Design

Systematic use of **models** throughout the development process



# Automatically generate code for prototyping and production




**Reliable** and **high performance**, with **flexible** choice of targets

Automation

Coding | Verification

# MathWorks supports deployment to a variety of low-cost hardware



Arduino



Lego EV3



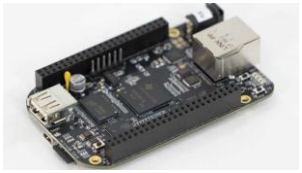
Raspberry Pi



Android/iOS Devices



Kinect for Windows



BeagleBone Black



Texas Instruments



STM Electronics



Freescale



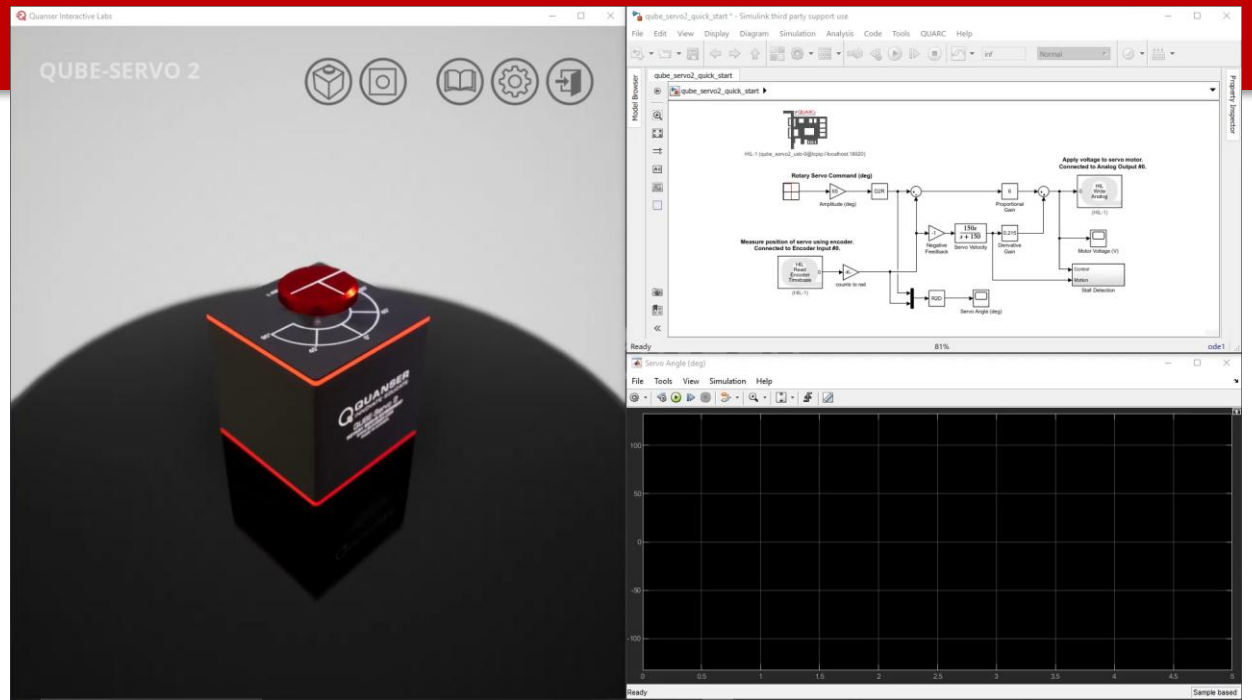
Zynq SDR

Automation

Coding | Verification

# MathWorks supports project-based learning through **Quanser virtual labs** and **hardware**

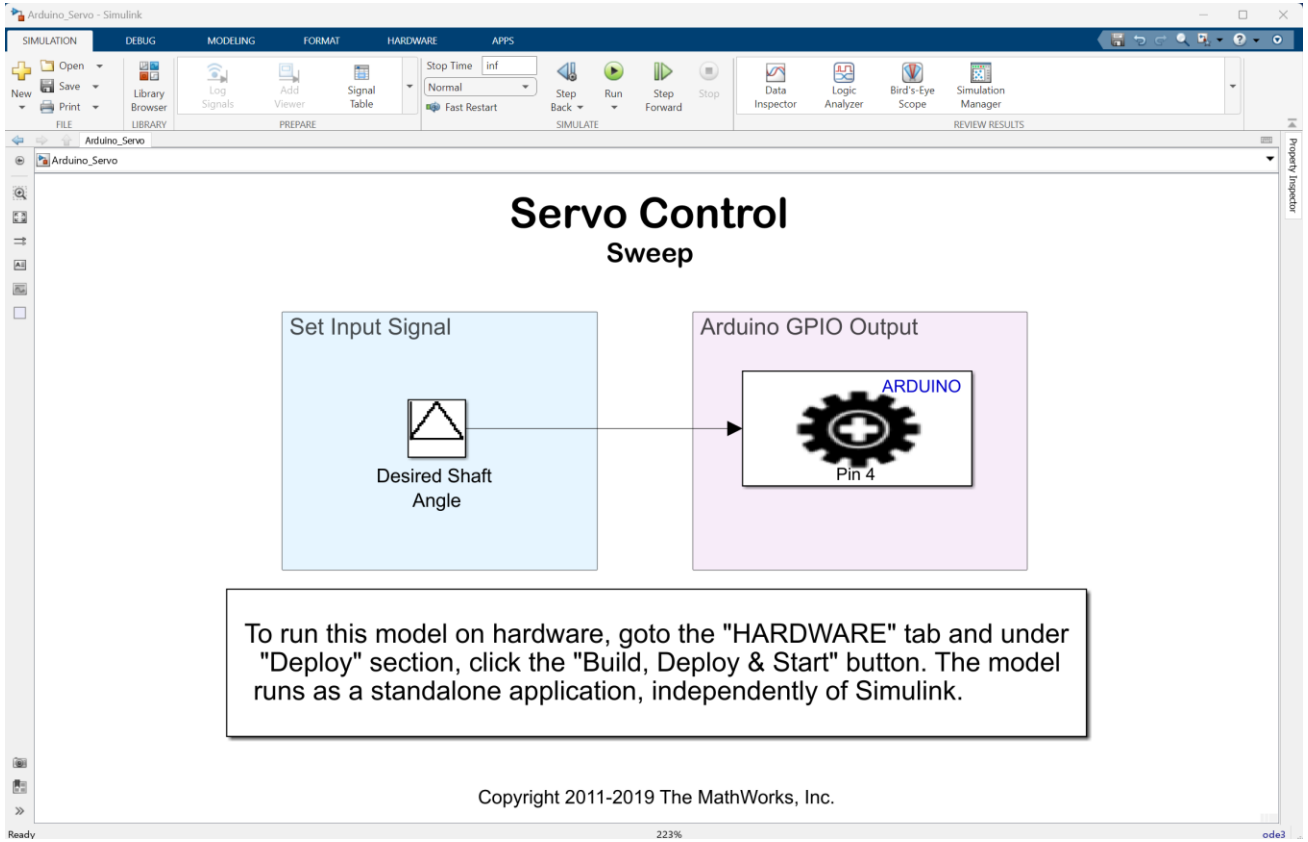
Quanser Interactive Labs is a hardware-inspired, **scalable** way to bring credible, high-fidelity interactive lab experiences to students in **blended, hybrid, or remote laboratory** configurations.






# Exercise 3: Control a servo with Arduino and Simulink

- Open: ***Arduino_Servo.slx***
- Add desired block as input
- Select output GPIO pin
- Run model with various settings:
  - ***Connected IO***
  - ***Monitor & Tune***
  - ***Build, Deploy, & Start***



# Agenda

- Introduction to Model-Based Design
- Simulink 101 – An Industry Standard tool for Model-Based Design
- **AI use in Industry and Safety Critical Systems**
- Resources

# Exploring AI workflows in Model-Based Design with MATLAB and Simulink – **Lucid Motors**



## As an Engineer, I want to ..

- Be efficient with my time on the project
- Feel accomplished while working on the project
- Write good documentation (But I don't want to write it myself 😐)

LUCID

# Lucid put an **AI agent** inside virtual vehicle development

Lucid's engineers build virtual vehicle models. In practice that meant constant switching between specs, scripts, models, and test results.

**Navigates the Simulink model**

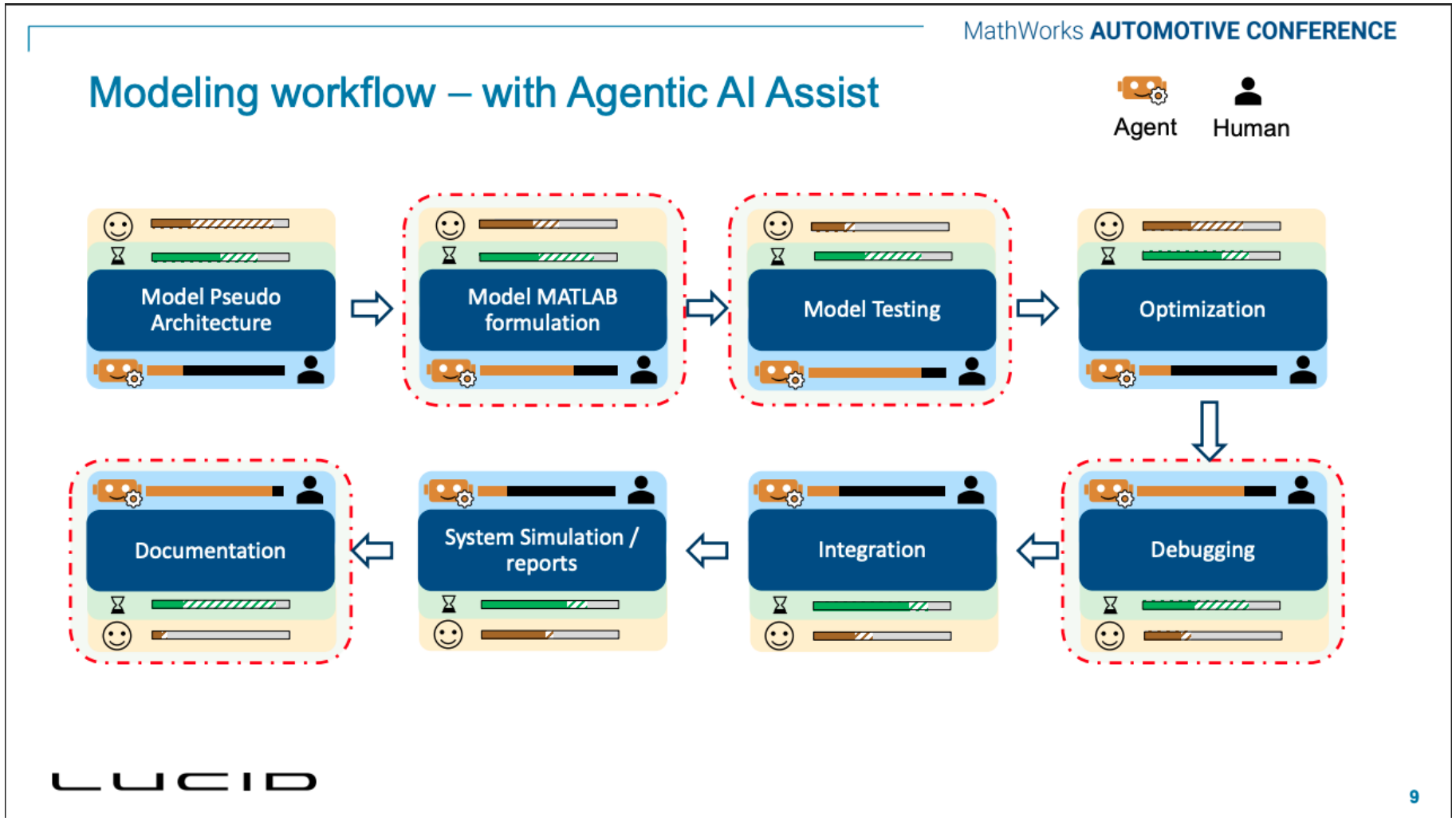
**Runs model-in-the-loop tests**

**Diagnoses errors, reports back**

**Engineers hold the decisions**

> *"Reduced integration time and rework, while maintaining traceability"*
>
> Fauzan Dhongre, Lucid. MathWorks Automotive Conference 2026.

# Exploring AI workflows in Model-Based Design with MATLAB and Simulink – Lucid Motors




MathWorks Automotive Conference North America 2026

# Let’s take a step back and look at three types of AI on a continuum

Predict → Generate → Plan & Act



## Discriminative AI

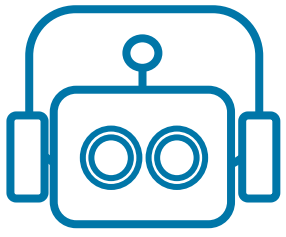
Map input to outputs to make predictions



## Generative AI

Generate new content from training data

Chat GPT · Claude · Google Gemini · GitHub Copilot



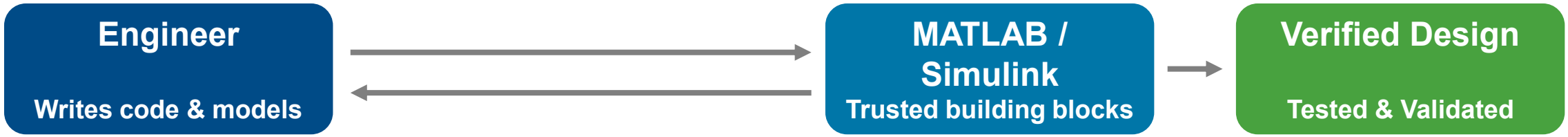
## Agentic AI

Plan and act using tools, memory, and trained logic

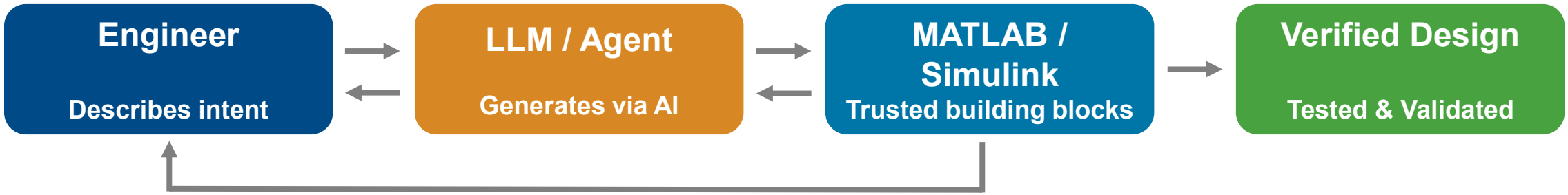
Gemini CLI · Open AI Codex · Claude Code

# AI changes creation speed, not engineering accountability

Today




With AI




*What doesn't change when you add AI*

**The engineer owns the design. The same engineering evidence verifies it.**

# Industry expects graduates will understand how to interface AI inside real systems

- How to **enforce requirements** in AI-supported development.
- How to **perform simulations** of AI-developed models before deployment.
- How to **test, verify, and validate** AI-supported system components.
- How to **integrate AI with physical systems** and real-time control systems.

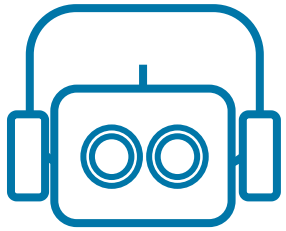
# What this means for engineering education

- AI literacy is now a fundamental skill
- **Engineering fundamentals including critical thinking are more important than ever**
- Focus shifts from creation → understanding and validation
- Responsible and effective AI use must be taught explicitly

# Accelerate your Simulink productivity with **GenAI assistant** and **Agentic workflows**



Generative AI



Agentic AI



**Simulink Copilot**

*optimized for Simulink*



**Simulink Agentic Toolkit**

*use Simulink with external AI Agents*

R2026a

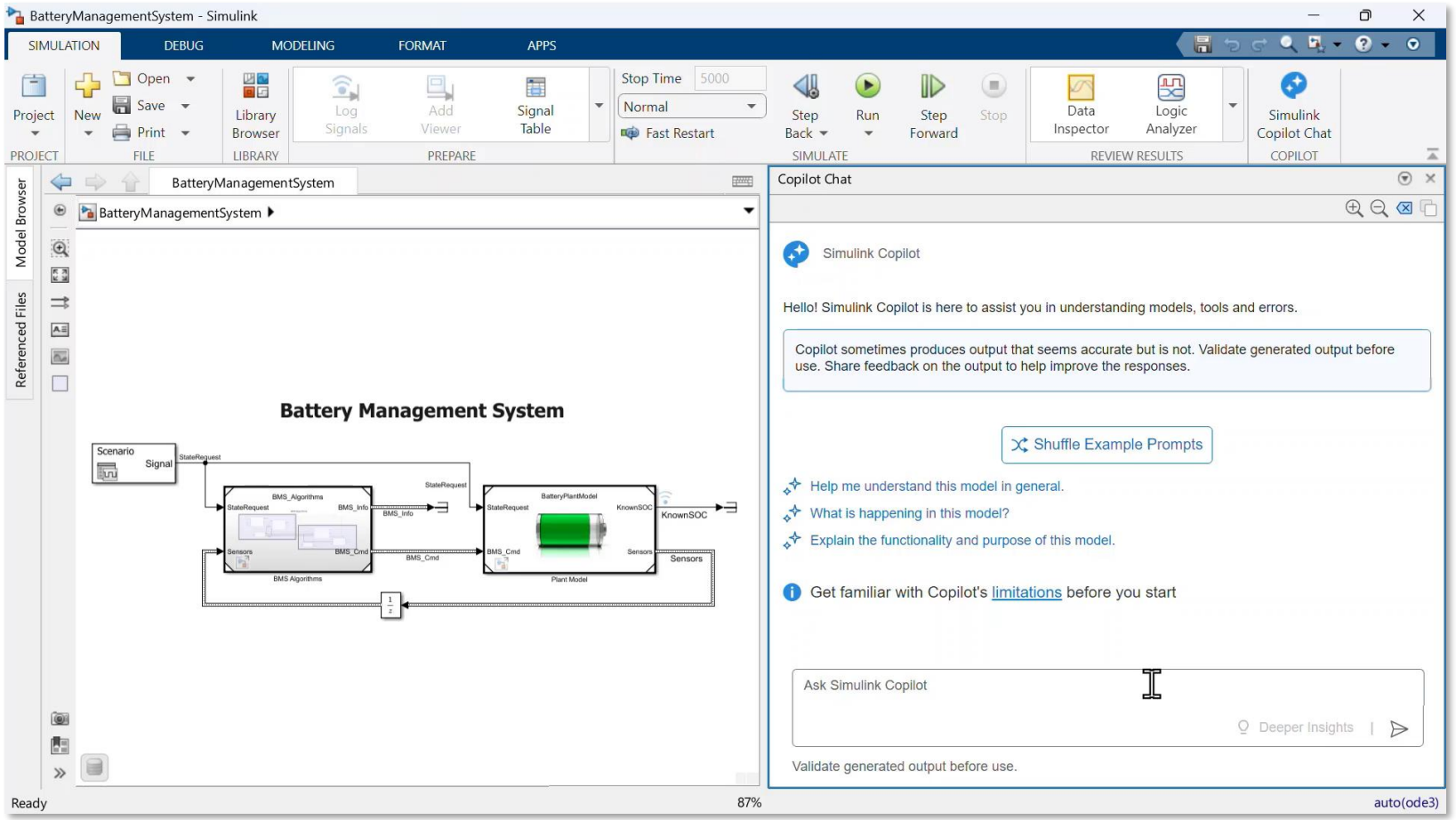
# Simulink Copilot
# Provides deep insights and actionable advice for Simulink models

## Explain & Advise

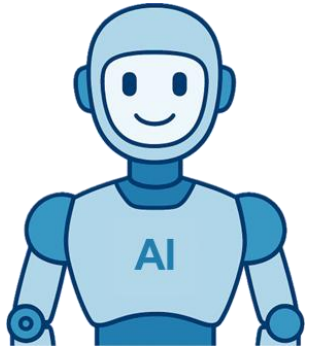
- **Learn tools & techniques**
- **Understand the model**
- **Search the model**
- **Get advice**
- **Troubleshoot errors**

## Review & Automate

- **Automate tasks**
- **Track progress**



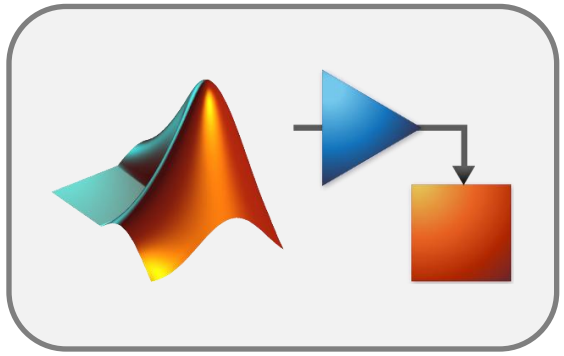
# Agentic AI workflows with MathWorks tools are enabled by Model Context Protocol (MCP)




MCP Client

MCP

MCP Server



**Cloud providers**
**OR**
**Local models**

Gemini CLI

Open AI Codex

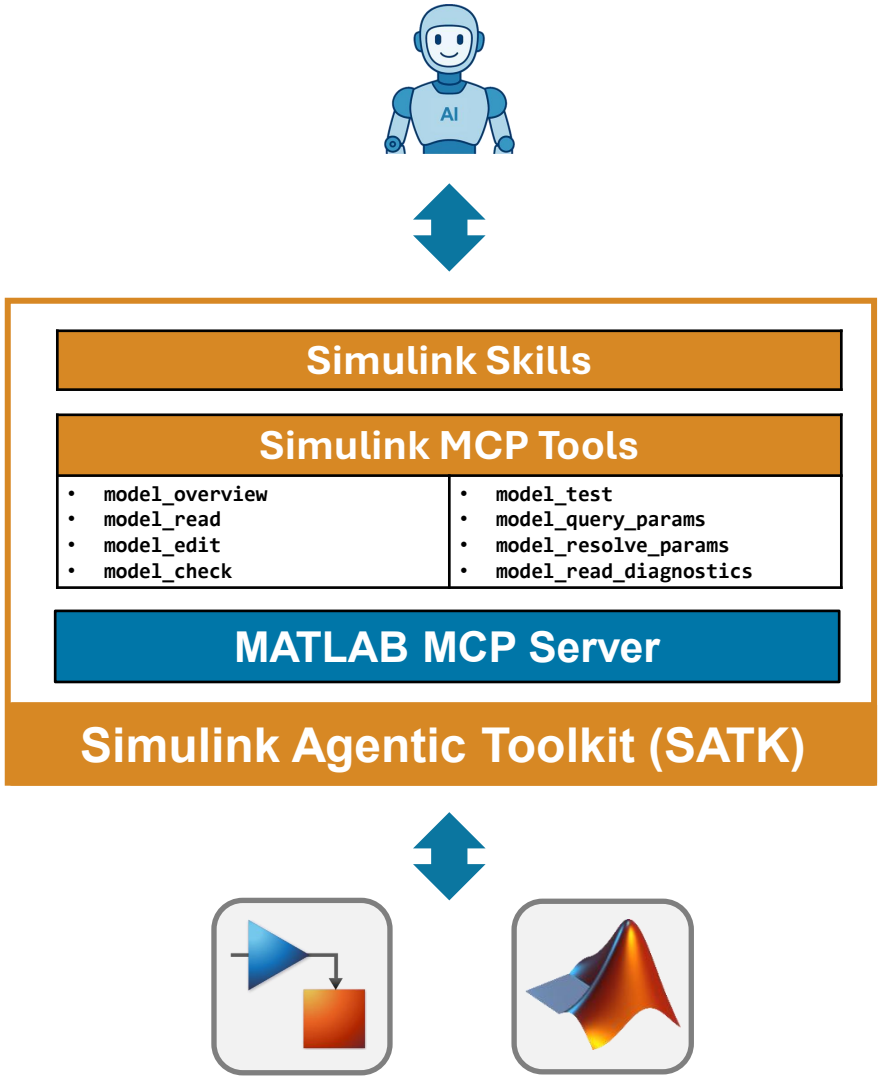
Claude Code

**MATLAB & Simulink**

# Simulink Agentic Toolkit

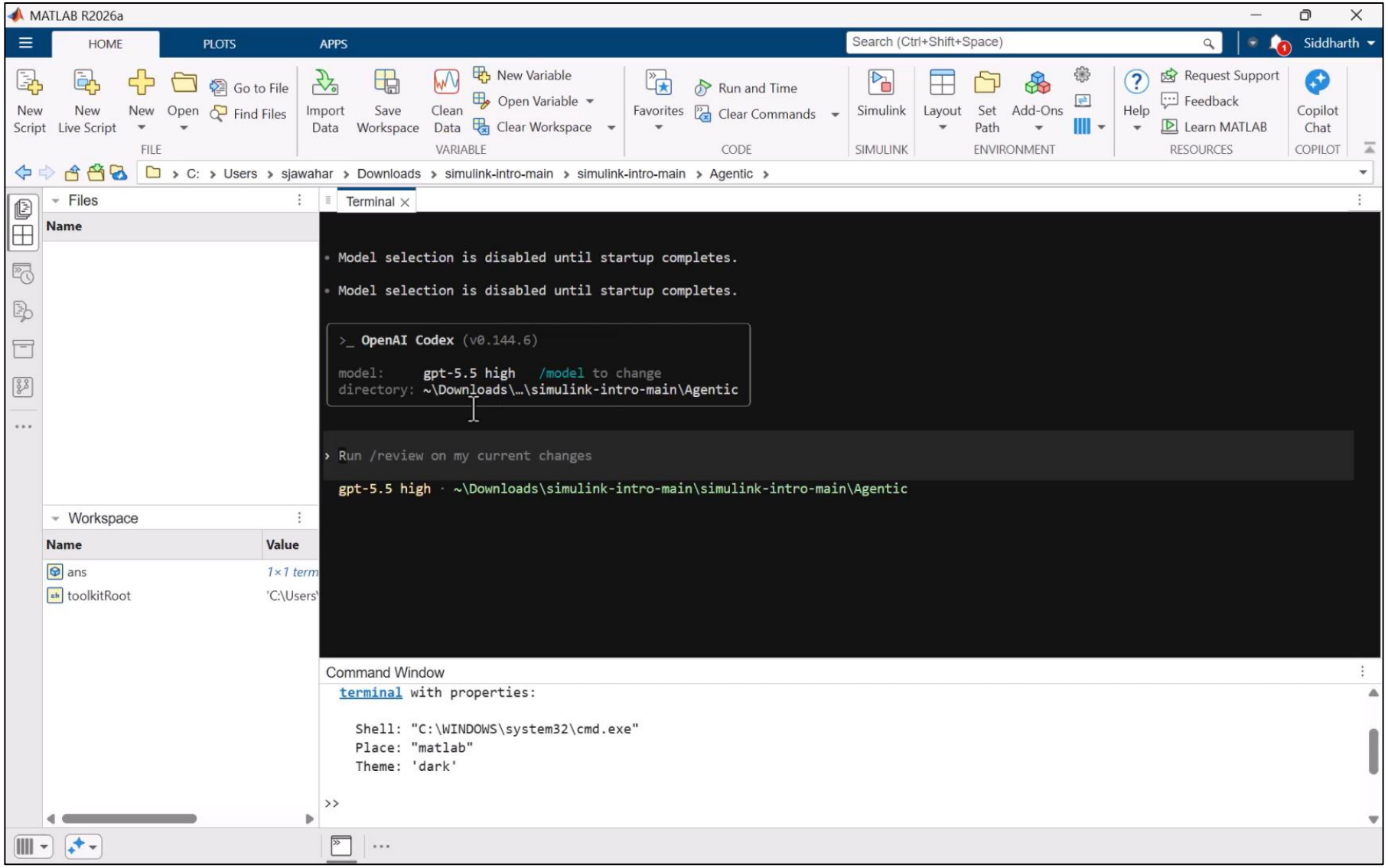
*Connect agents to Simulink giving them the ability and the knowledge to work with Simulink models effectively.*

- MCP tools optimized for reading, editing, querying, testing, and checking Simulink models
- Agent skills encoding Model Based Design best practices for model building, plant specification, testing, requirements, and architecture modeling




https://github.com/matlab/simulink-agentic-toolkit

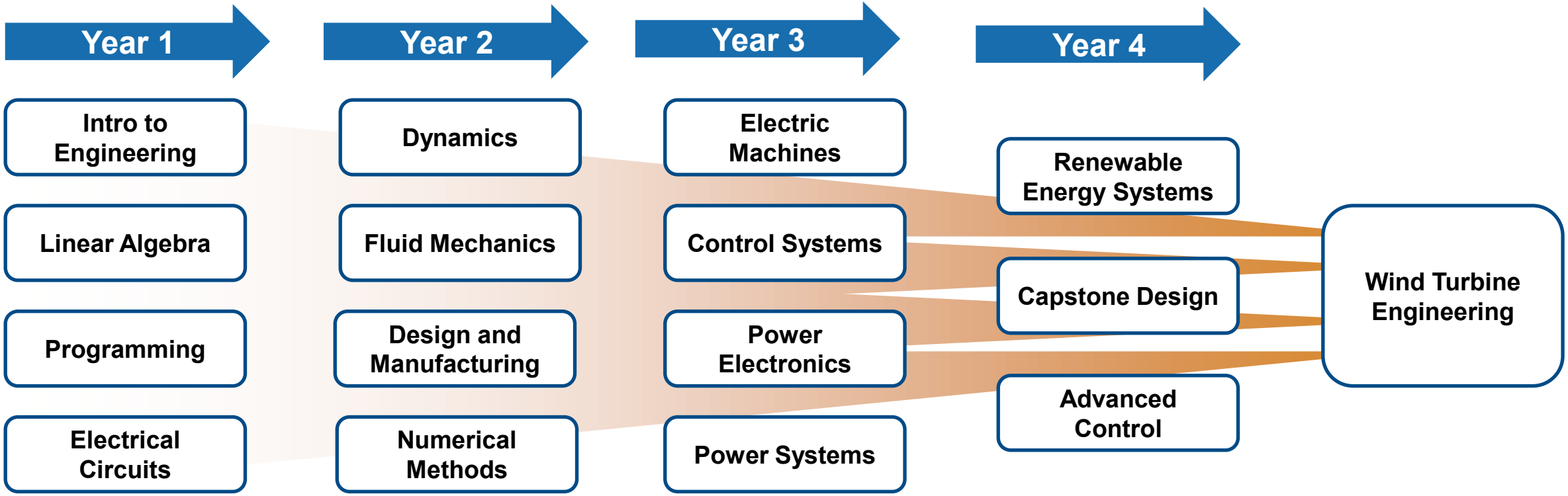
# Agentic Engineering in action – Mass Spring Damper + PID Controller



# Agenda

- Introduction to Model-Based Design
- Simulink 101 – An Industry Standard tool for Model-Based Design
- Accelerating your work with GenAI and Agentic AI
- **Resources**

# Introduce Simulink as a modeling tool from foundational courses

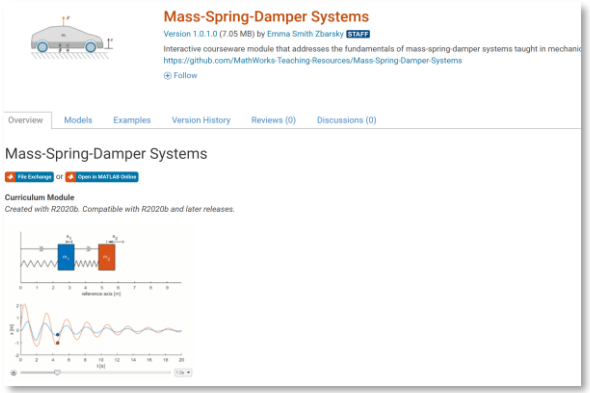



Fundamental Understandings & Skills

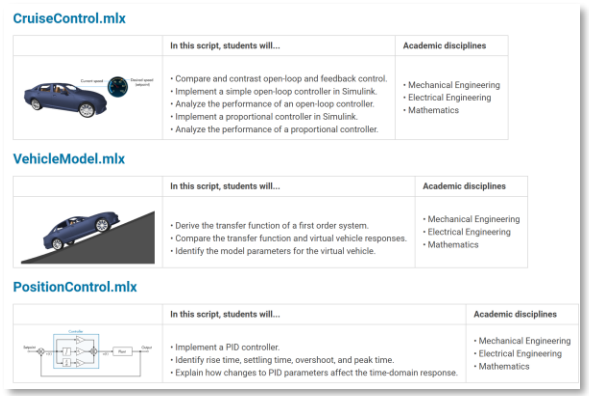
Core Engineering Topics & Subsystem Design

System Design & Implementation

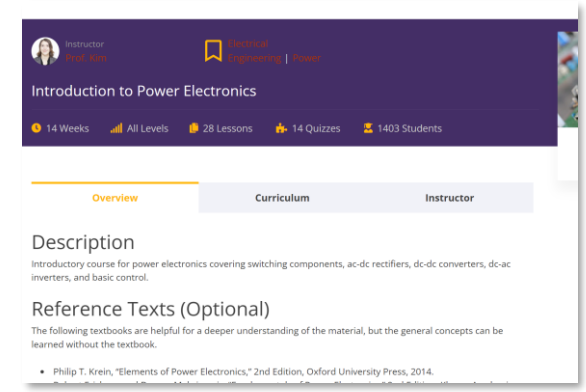
# Modular courseware and virtual labs



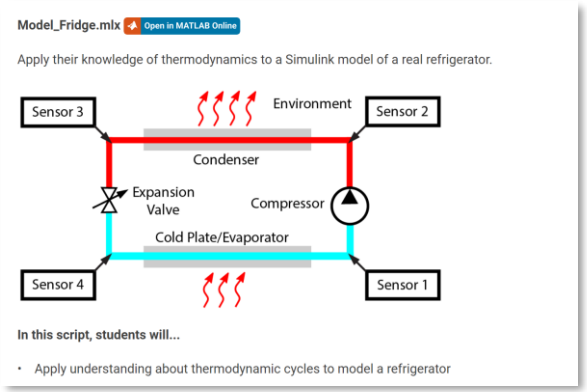

[Mass-Spring-Damper Systems]



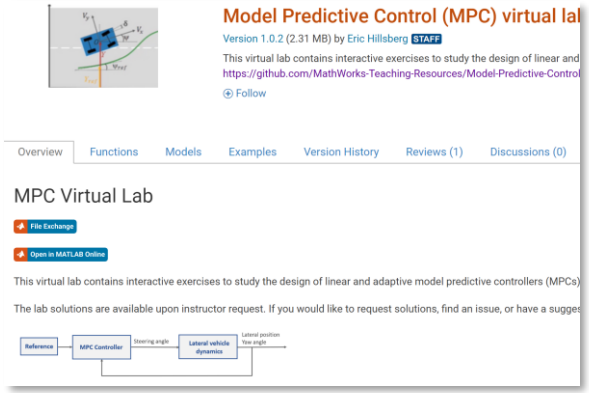

[Virtual Hardware and Labs for Controls]



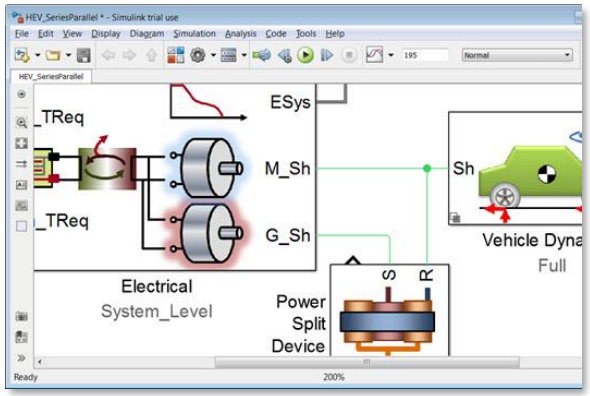

[Introduction to Power Electronics]




[Thermodynamics]

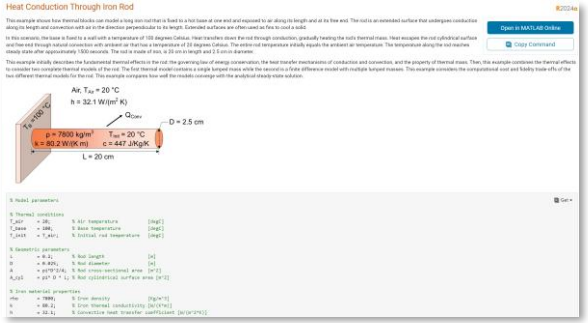



[Model Predictive Control]

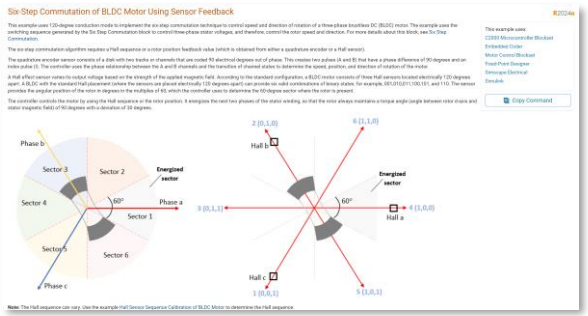



[Hybrid-Electric Vehicle Model in Simulink] & [Simscape]

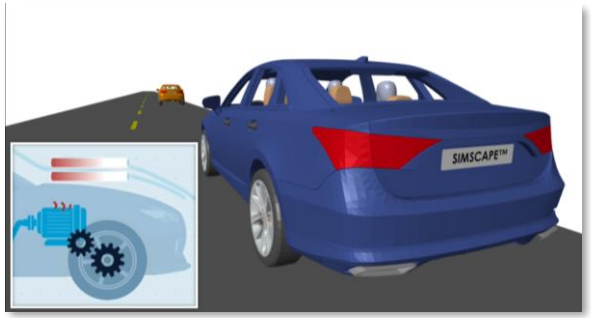
# Reference examples in documentation and file exchange



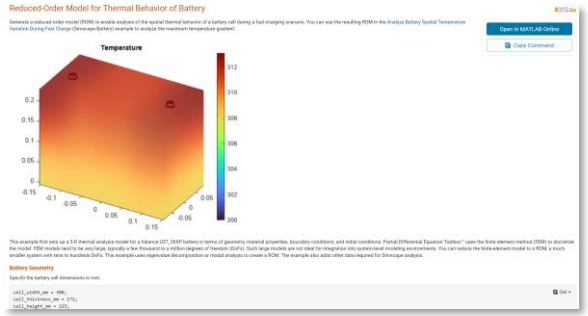

Heat Conduction Through Iron Rod



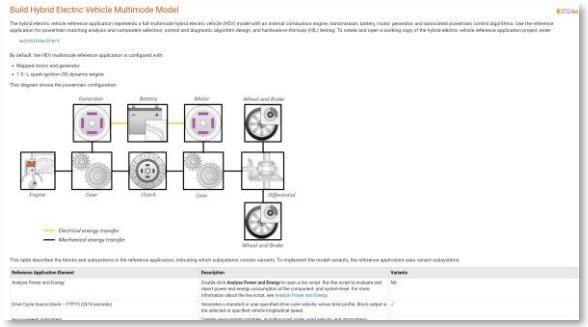

Six-Step Commutation of BLDC Motor Using Sensor Feedback



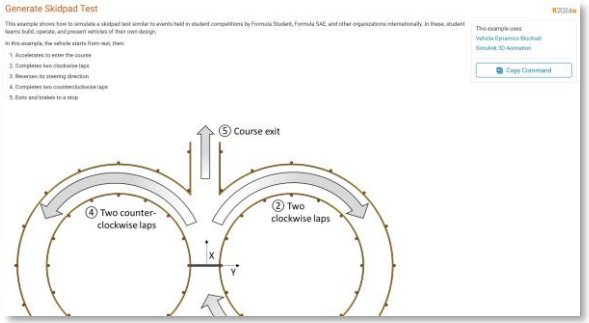

Battery Electric Vehicle with Motor Cooling in Simscape




Reduced-Order Model for Thermal Behavior of Battery




Build Hybrid Electric Vehicle Multimode Model

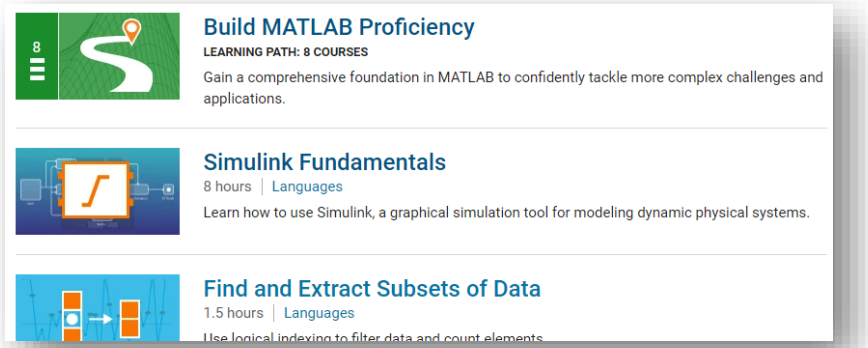



Generate Skidpad Test

Examples with Extended Descriptions and Educational Materials

# Self-Paced Online Courses

http://matlabacademy.mathworks.com




- MATLAB and Simulink online training courses at 44 Short Courses
  - 12 Learning Paths
- Offers learning flexibility for:
  - upskilling/reskilling
  - exploring new topics and ideas
  - meeting prerequisites for instructor-led courses or consulting engagements
- Choose to take individual lessons, short courses, or full learning paths
- Wide range of available topics—new courses and topics added every few months

## Short courses (1-3 hours)

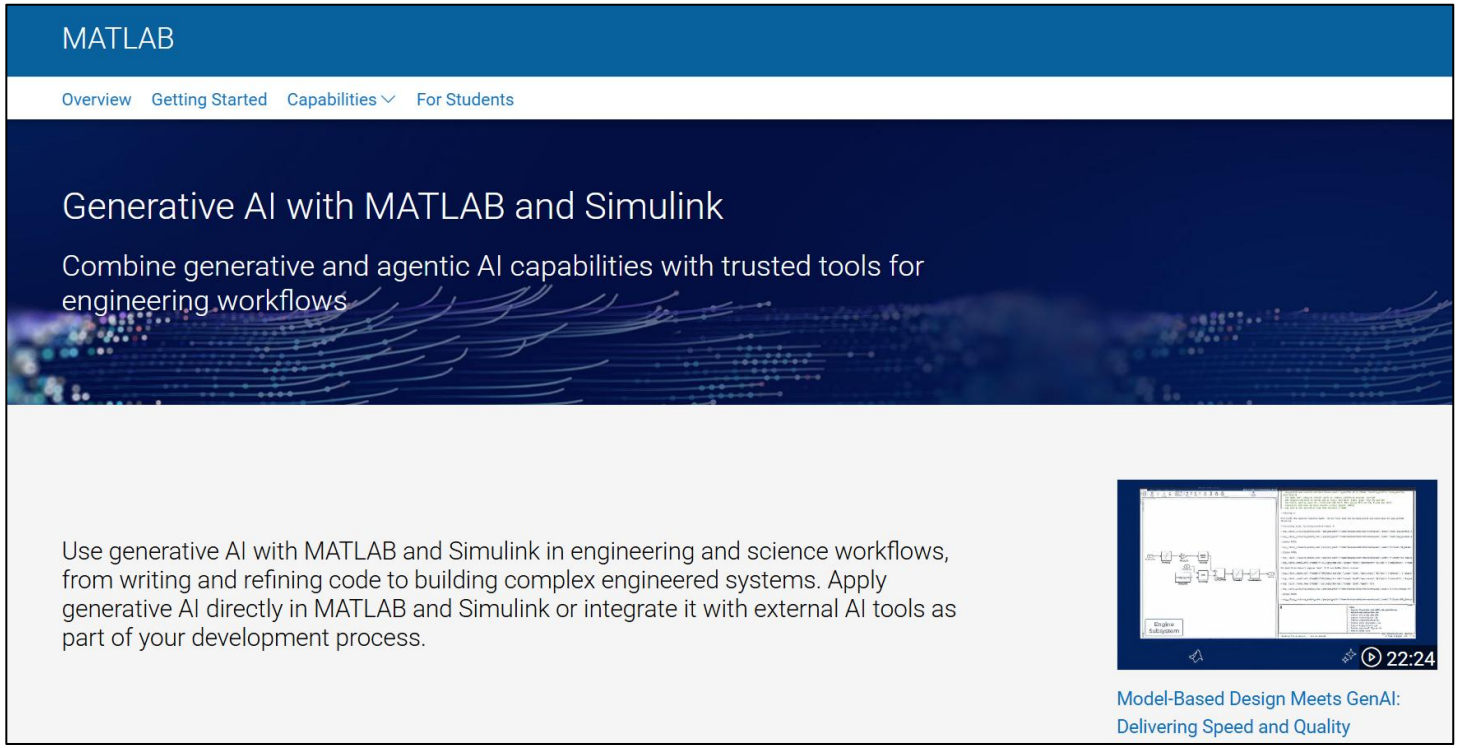
- MATLAB Onramp
- Simulink Onramp
- Circuit Simulation Onramp
- Machine Learning Onramp
- Power Systems Simulation Onramp
- Image Processing Onramp
- System Composer Onramp
- Object Oriented Programming Onramp
- App Building Onramp
- Simscape Onramp
- Stateflow Onramp

## Long courses (7-10 hours)

- Simulink Fundamentals
- Image Processing with MATLAB
- Deep Learning with MATLAB
- Machine Learning with MATLAB

# GenAI and Agentic AI offerings from MathWorks

- GenAI assistants
  - MATLAB Copilot
  - Simulink Copilot
- Agentic AI
  - MATLAB Agentic Toolkit
  - Simulink Agentic Toolkit



# MathWorks Education Application Engineer *sjawahar@mathworks.com*

consult with faculty and researchers to support them with their STEM initiatives,

including integrating computational or systems thinking into their curriculum and research









# Take a min to submit a survey!

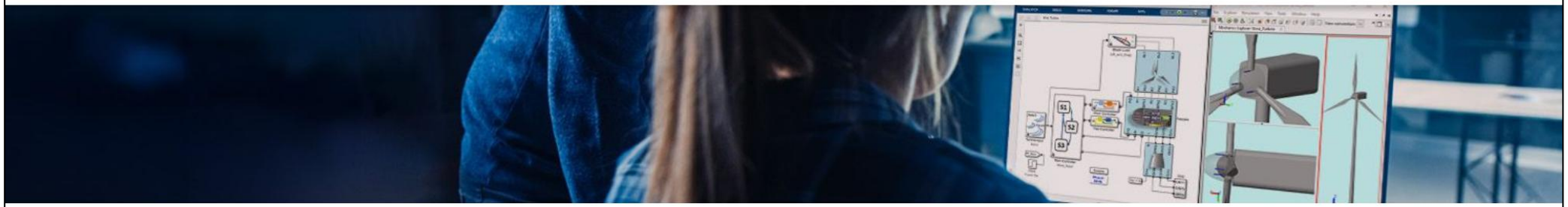


[Attendee Survey - Building Industry-Ready Skills with MBD, Simulink, and Agentic AI – Fill out form]

MathWorks

MathWorks MATLAB

Live Events



# From Control Theory to Practice: Addressing Misconceptions in Control Design

| Start Time | End Time |
| --- | --- |
| 20 Oct 2026, 1:00 PM EDT | 20 Oct 2026, 2:30 PM EDT |

Share Post

## Overview

Controls courses provide a strong foundation in system dynamics, stability, and controller design. However, when control theory is applied to real engineering problems, additional challenges quickly emerge, including uncertain plant models, hardware constraints, integration with other system components, verification, and deployment.

These realities can challenge common assumptions held by students and early-career engineers. For example, they may assume that the best controller is simply the one with the highest performance, or that most engineering effort is spent on control design alone. In practice, engineers must balance performance with

**Work or university email***

sjawahar@mathworks.com

I'm interested in product pricing or a trial

**Please select the seminar you would like to attend:***

20 Oct 2026, 1:00 PM EDT

*Required field

Link